Multinational Responses to Coordinated Tax Reform: Evidence from the Global Minimum Tax

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Abstract

We study how multinational enterprises (MNEs) respond to an internationally coordinated change in corporate taxation. The introduction of the Global Minimum Tax (GMT) in 2024 subjected MNEs with revenues above EUR 750m to a minimum effective rate of 15%. We exploit this revenue threshold in a difference-in-differences framework to assess the impact of the GMT on effective tax rates, investment and employment. Using group-level Orbis data, we find that previously low-taxed MNEs increased effective tax rates by around 1.7 percentage points, and increased global corporate tax revenues by 2.4-3.4%. However, this increase was not accompanied by a detectable reduction in investment or employment. The increase in effective tax rates is concentrated in MNEs more likely to have greater profit shifting opportunities and higher profitability. The findings provide first ex-post evidence on the effects of the GMT and suggest that internationally coordinated reforms can raise the effective taxation of MNEs without generating short-run reductions in economic activity.

Keywords: Corporate taxation, international tax cooperation, multinational enterprises, global minimum tax, investment

JEL Codes: H25, H32, E22, F21, F23

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1 Introduction

The effect of corporate taxation on real economic activity, a central question in public economics, may importantly depend on the extent to which tax policy is coordinated across countries. A unilateral tax increase by a jurisdiction can induce multinational enterprises (MNEs) to relocate profits, investment or other economic activity to other jurisdictions. These responses can in turn weaken governments' ability to raise revenue from corporate income taxation and are seen as a contributor to a long-run decline in statutory corporate tax rates observed across countries. A coordinated tax reform, on the other hand, may limit opportunities to shift activity across jurisdictions.

The introduction of the Global Minimum Tax (GMT) constitutes the first large-scale attempt at an internationally coordinated reform to corporate taxation which may mitigate such responses. The reform was agreed by around 140 member jurisdictions of the Inclusive Framework on BEPS and is designed to ensure that large MNEs with revenues above EUR 750 million are subject to a 15% effective minimum tax rate on their excess profits regardless of where they operate (OECD, 2021b). Unlike conventional corporate tax reforms that change tax rates in a particular jurisdiction, the GMT changes the international tax environment faced by large MNEs across all jurisdictions in which they operate. The reform has generated an intense debate about its likely effects on the corporate tax burden of MNEs, their investment, employment and profit-shifting behaviour. This debate, however, has been informed almost exclusively by *ex ante* modelling and simulations.

We provide the first *ex post* evidence on how MNEs respond to the globally coordinated change in corporate taxation brought about by the GMT. Specifically, we estimate the observed responses of MNEs to the announcement of the GMT in 2021 and its introduction from January 2024. As a first step, we establish whether its announcement and implementation affected MNEs' effective tax rates (ETRs). Second, we test if such changes translate into reduced investment or employment at the MNE group level. Identification exploits the quasi-experimental variation generated by the EUR 750 million revenue threshold determining GMT scope. Using a difference-in-differences (DiD) approach, we compare firms which are relatively more likely to be affected by the GMT (the treatment group) to those which are relatively less likely to be affected (the control group). For this, we draw on consolidated financial statements and historical information on company ownership links from the Orbis database.

The results suggest that exposure to the GMT increased ETRs of MNE groups subject to the reform, but did not impact investment or employment in the short run. The analysis does not point to announcement effects for any of the outcome variables. After the intro-

duction of the GMT, however, MNEs affected by the reform experienced an increase in their consolidated ETRs of between 1 and 2 percentage points (ppts) relative to the control group. MNEs with relatively low pre-reform ETRs, which are likely to be most affected by the reform experienced a statistically significant increase in their effective tax rates of 1.7 ppts. For all in-scope MNEs the estimated increase is 1.4 ppts. Despite the increase in effective taxation, there is no evidence of reductions in investment or employment in the first post-implementation year. The analysis also shows no impact on MNE organisational structures in the short run, such as the number of subsidiaries or presence in investment hubs. Heterogeneity analysis further suggests that the effects on consolidated ETRs are primarily driven by MNEs with characteristics suggesting greater propensity for higher tax aggressiveness and by more profitable MNEs. The point estimates imply an additional EUR 79bn-109bn of CIT revenue globally in the first year of the GMT's implementation, representing an increase in total global CIT revenue of 2.4-3.4%.

The paper makes two key contributions. First, it contributes to the literature on international tax coordination by providing the first causal evidence on realised MNE responses to a large-scale coordinated corporate tax reform. Much of the theoretical literature argues that non-cooperative tax setting may lead governments to choose inefficiently low levels of corporate taxation (Kanbur and Keen, 1993; Johannesen, 2022). Papers estimating the real impacts of corporate taxation on investment have traditionally focused on country-level CIT reforms (Chodorow-Reich et al., 2024b; Desai et al., 2004; Zwick and Mahon, 2017; Harju et al., 2022), leaving open the possibility of MNEs shifting across borders to avoid the effects of unilateral tax changes. Some more recent papers have considered spillovers of CIT reforms on MNE activities in other jurisdictions (Hanappi et al., 2023; Hanappi and Whyman, 2023; De Vito et al., 2025), while others have considered the impact of the international tax environment on the location of profits and assets (Keen et al., 2023; Chodorow-Reich et al., 2024a; Bilicka et al., 2024). The GMT provides a unique opportunity to assess whether international coordination can mitigate the cross-border responses that may undermine unilateral corporate tax reforms, including profit shifting and the relocation of economic activity. The findings suggest that coordinated minimum taxation can increase effective tax rates and tax revenues without detectable short-run reductions in investment or employment.

Second, the paper contributes to the growing literature on the GMT itself. Most empirical studies focus on *ex ante* estimates of corporate tax revenue effects (Devereux et al., 2020; Baraké et al., 2021; IMF, 2023; Joint Committee on Taxation, 2023; Hugger et al., 2024). Additional theoretical papers also study revenue impacts and their distribution across country groups (Janeba and Schjelderup, 2023; Haufler et al., 2025). Others study the potential MNE responses to the GMT such as expected reductions in profit shifting (Johannesen, 2022;

Ferrari et al., 2023; Bratta et al., 2024; Fuest et al., 2025). Several papers investigate how the GMT might impact investment (OECD, 2020; UNCTAD, 2022), with some highlighting the potential positive impacts on investment outside of low tax jurisdictions (Keen et al., 2023), and others focusing on the potential negative impact on investment, especially for intangible intensive firms (Bilicka et al., 2024; Dyrda et al., 2026). These estimates necessarily rely on assumptions regarding the extent of jurisdictions' implementation, extrapolations of pre-implementation data, and are subject to considerable uncertainty regarding MNE behavioural responses. This paper is the first to use post-implementation data to assess whether the central predictions of the literature have materialised in practice.

The rest of the paper proceeds as follows. It begins with a description of the institutional setting in Section 2, followed by a discussion of possible responses to the GMT based on the existing literature in Section 3. Section 4 provides a description of the data used for the analysis and the empirical methodology. Section 5 describes the main results, as well as heterogeneity and robustness checks. In Section 6, the estimation results are applied to provide an estimate of the revenue impacts of the reform. Section 7 concludes.

2 Institutional setting

The GMT introduces a coordinated change to the taxation of large MNEs through a series of interlocking mechanisms. It is governed by the Global Anti-Base Erosion (GloBE) Rules, agreed in October 2021 by around 140 jurisdictions which make up the OECD/G20 Inclusive Framework on Base Erosion and Profit Shifting. While the participating jurisdictions are not obliged to implement the rules, they commit to consistent application of the rules if they choose to do so and accept the application of the rules by other members. The GloBE rules aim to ensure that large MNE groups are subject to a minimum ETR of at least 15% on their excess profits in every jurisdiction they operate. Excess profits are defined as those profits exceeding a profitability threshold defined as a fraction of assets and payroll. The rules are set out in detail in OECD (2021b), complemented by a Commentary and Administrative Guidance that has been updated several times (OECD, 2022, 2023b). An update in early 2026 introduced simplifications and permanent safe harbours which will be incorporated into the Commentary (OECD, 2026). This section provides a high-level overview of the GloBE Rules focusing on the scope of the rules, the calculation of the top-up taxes, and the rule order in collecting top-up taxes. The section also summarises the timeline leading to the implementation of the GMT starting in 2024, with a view to informing the identification of the potential effects of the GMT in this paper.

2.1 Overview of the GloBE rules

The GMT applies to MNE groups with annual revenues of EUR 750 million or more, with certain exclusions. Specifically, companies with more than EUR 750m in revenues in two of the four preceding years and with at least one foreign subsidiary are generally in scope of the GMT. There exists a small number of exclusions including for Governmental Entities, International Organisations, Non-profit Organisations, Pension Funds, and MNE groups in the initial phase of their international activity. Additional exclusions apply to income from international shipping activities as well as for Investment Funds and Real Estate Investment Vehicles if they are the Ultimate Parent Entity (UPE) of the MNE group. Commencing in 2026, MNE Groups subject to the Side-by-Side Safe Harbour discussed below are also excluded from certain parts of the system.

The GMT applies top-up taxes based on ETRs calculated at the level of MNE subgroups, comprising all entities of the MNE group located in a jurisdiction. Where an in-scope MNE profit is subject to an ETR below the 15% rate in any jurisdiction where it is operating, a top-up tax equal to the difference between the 15% rate and the ETR in the jurisdiction can be levied (see Figure 1). Since tax bases differ across jurisdictions, the GloBE rules rely on financial accounts to build a consistent approach to calculating the ETR for each MNE subgroup. This "GloBE ETR" is computed by dividing all covered taxes of entities located in the jurisdiction by their total GloBE Income. Covered taxes are based on annual tax expense of all entities in a subgroup of in-scope MNEs. Similarly, GloBE Income combines all profits and losses of the entities in a subgroup. Both are calculated by applying certain agreed adjustments to the financial accounting figures to address certain book-tax differences.[1] Common causes for such adjustments are e.g. prior-period losses, provisions that allow for accelerated (tax) depreciation, or certain tax incentives that qualify for preferential treatment under the GloBE rules. In addition, the tax base is reduced by the Substance-Based Income Exclusion (SBIE), which excludes a 'routine' return from top-up taxation. The SBIE is based on the amount of tangible assets and payroll reported in a jurisdiction. After a transitional period, the SBIE equals the sum of 5% of the value of tangible assets and 5% of the value of payroll.

[Figure 1 here]

Top-up taxes due under the GMT are distributed across jurisdictions based on an agreed rule order. Figure 2 shows the order of top-up tax collection under the GloBE rules. If

[1] If GloBE income is negative for a given year, no top-up tax will be levied in this year.

top-up tax is levied on the profit an MNE reports in a jurisdiction, that jurisdiction can itself collect the top-up tax in priority to other jurisdictions, by implementing a Qualified Domestic Minimum Top-up Tax (QDMTT, Panel 1 of the Figure). Any top-up tax remaining after the application of QDMTTs can be collected by the UPE jurisdiction under the Income Inclusion Rule (IIR, Panel 2). If the IIR has not been implemented by the UPE jurisdiction, top-up taxes can also be collected under the IIR by an intermediate parent entity (IPE, Panel 3). Finally, if any top-up tax remains, it can be imposed by multiple jurisdictions under the UTPR (Panel 4). The UTPR is a backstop ensuring that all low-taxed income is captured under the GMT. Each jurisdiction which has adopted the UTPR can collect top-up tax in proportion to the amount of substance of the MNE group in the jurisdiction, where substance is measured by tangible assets and employees at equal weights. These interlocking rules mean that the top-up tax collected by a given jurisdiction depends on the actions of other jurisdictions.

The Side-by-Side package agreed in 2026 adds different treatment of MNEs headquartered in jurisdictions with tax regimes already incorporating a system of minimum taxation. These pre-existing systems need to achieve similar outcomes as the GMT. MNEs falling under the Side-by-Side Safe Harbour are not subject to the IIR or UTPR. Where subsidiaries are located in QDMTT jurisdictions, however, MNEs falling under the Safe Harbour will still be subject to those QDMTTs. This treatment commences in 2026. As of January 2026, the only jurisdiction with a tax regime qualifying for this safe harbour is the United States which have benefited from a temporary safe-harbour before the agreement on the Side-by-Side package. This implies that the GMT only changes the taxation of US MNEs compared to a pre-GMT situation to the extent that subgroups of these MNEs operate in jurisdictions that have implemented a QDMTT.

[Figure 2 here]

2.2 Timeline of GMT introduction

The development of the GMT involved several distinct stages, with a delay between announcement and implementation opening the possibility of separate announcement and implementation effects. The GMT was introduced under the auspices of the OECD/G20 Inclusive Framework on Base Erosion and Profit Shifting (BEPS). Key steps towards the announcement and implementation of the GMT are summarised in Figure 3. The BEPS project was initiated in 2013 to address the erosion of corporate tax bases through corporate

tax avoidance and evasion. The Final Reports published in 2015 listed 15 Actions that aimed to reduce tax avoidance opportunities of multinational companies.[2] In 2016, the Inclusive Framework on BEPS was established inviting all jurisdictions to participate and further develop the work on BEPS.

Discussion on a GMT began as part of a second phase of the work of the Inclusive Framework developing a Two Pillar Solution to Address the Tax Challenges of the Digital Economy. A first discussion paper from 2019 included suggestions about potential revenue-based scoping thresholds with a blueprint report providing additional detail (OECD, 2020). In October 2021, the Inclusive Framework on BEPS formally approved an international tax agreement stemming from these negotiations (OECD, 2021a). This outcome statement outlined the design of the GloBE rules which implement the GMT, including the revenue threshold of EUR 750m used for identification in this paper. This was followed by the publication of more detailed model rules two months later. This credible announcement of the GMT is used in this paper to test for announcement responses starting in 2022. A key first step towards the implementation of the GMT was the approval of the Council Directive (EU) 2022/2523 in December 2022 which formalised the implementation in the EU member states. The directive introduced the GloBE rules for fiscal years starting in or after January 2024. Similar steps were taken by several other large economies. Some jurisdictions also chose to only implement parts of the GloBE rules, such as the QDMTT. As of today, more than 50 jurisdictions have taken steps towards implementation of the GMT. Given its design, the implementation in the EU and many other large economies including Australia, Canada, Japan, Korea and the UK, as well as several major investment hubs imply that a high share of large MNEs will be subject to the GMT. [3]

[Figure 3 here]

3 Expected impacts

The proposed empirical approach first estimates whether the GMT increased the ETRs of affected MNEs and proceeds to estimate changes in other outcomes as a second step. This

[2]One of these Actions, Action 13 introduced Country-by-Country reporting for multinationals earning revenue above EUR 750m to improve transparency on their global activities for tax authorities. This reporting requirement started to apply in certain jurisdictions from 2016. The GMT subsequently adopted a modified version of this threshold.

[3]See https://www.oecd.org/en/topics/sub-issues/global-minimum-tax/central-record-of-legislation-with-transitional-qualified-status.html for the central record on purposes of the GMT, listing all jurisdictions that have implemented IIRs and QDMTTs.

reflects the expectation that increases in corporate tax rates are necessary for other MNE responses to be observed. A number of existing papers have predicted that the GMT would have a quantitatively significant impact on tax revenues or ETRs of MNEs based on ex-ante analysis (Devereux et al., 2020; Baraké et al., 2021; IMF, 2023; Joint Committee on Taxation, 2023; Hugger et al., 2024; Janeba and Schjelderup, 2023). However, given the uncertainty surrounding these ex-ante estimates, the estimation of the impact on MNE ETRs in the first year following GMT implementation is an important test of whether the predictions have been borne out in practice.

The ETR in this paper is a backward-looking measure of the effective average tax rates. It is measured as the ratio of reported taxes at the consolidated level in a given fiscal year to the corresponding measure of pre-tax profit. Although many papers focus on forward-looking effective marginal or average tax rates as being relevant to firm investment decisions, such forward-looking rates are difficult to measure at the MNE group level (see e.g., Devereux and Griffith (2003)) for an early contribution). In addition, companies in practice often rely on measures such as average book-ETRs as a heuristic to also evaluate incremental investment decisions according to survey-based evidence (Graham et al., 2017). This implies that the backward-looking ETR used in this paper is likely to be a good proxy for the tax rates relevant for investment decisions.

The academic literature suggests that corporate tax rates may impact investment through two channels. Firstly, the neoclassical model of investment behaviour predicts that corporate tax rates reduce investment by increasing the user cost of capital (Hall and Jorgenson, 1967). Several studies have predicted that the increased rates of taxation associated with the GMT will result in reductions in real investment based on this channel (Keen et al., 2023; Bilicka et al., 2024). Alternatively, increases in corporate tax rates can reduce investment by tightening companies' financial constraints (Dobbins and Jacob, 2016; Egger et al., 2020).[4] However, the magnitude of investment responses to top-up taxation may importantly depend on the nature and extent of profit-shifting behaviours of MNEs (Juranek et al., 2023; Clifford et al., 2025). For instance, if low-tax outcomes are associated with infra-marginal profit-shifting behaviour, investments may not be significantly affected (Juranek et al., 2023). Additionally, the SBIE reduces the tax base in proportion to tangible assets and payroll and may therefore mitigate the increase in the marginal cost of real activity. Dyrda et al. (2026) formalise this mechanism in a quantitative model of multinational production, showing that the SBIE can lower the user cost of tangible capital in low-tax affiliates. At the same time, they show that GMT reduces the return to profit shifting and hence to intangible investment, creating a

[4]At the same time, this channel might be more likely to affect younger and smaller firms, so may be of limited relevance for large MNEs.

potentially countervailing channel for real activity.

An important distinction exists between investment responses measured at the level of the entire MNE group and responses measured at the level of entities of an MNE group in each jurisdiction. Since top-up taxation under the GMT applies based on ETRs measured at the jurisdictional level, a degree of reallocation of investment and employment may be expected within MNE groups. Thus, for example, potentially considerable investment responses may be observed at the level of particularly low-taxed jurisdictions, group-level responses may be much more muted. Some papers have found empirically that certain forms of profit shifting result in investments being reallocated between affiliates across jurisdictions, and do not find that restricting profit-shifting opportunities leads to significant changes in investment at the level of MNE groups (de Mooij and Liu, 2020; Knoll et al., 2021). Finally, many of the empirical studies have relied on tax reforms in one jurisdiction, leaving tax rates in other jurisdictions unchanged (Zwick and Mahon, 2017; Maffini et al., 2019). Given the unprecedented nature of the coordinated change in MNE taxation, the existing literature studying the effects of corporate taxation on investment might not apply in the context of the GMT.

Overall, there are several key scenarios for what may be expected in the first and second estimation steps. These scenarios are summarised in Table 1, separately presenting the expected effects on ETRs, investment and employment.

- Scenario (a), described in the first row, consists of the GMT not having any impact on MNEs' ETRs. Such a situation may arise if MNEs either have ETRs above the 15% minimum floor in all jurisdictions in which they operate, if they have sufficient substance (in the form of payroll and tangible assets) to protect any low-tax profits under the GMT's substance carveouts (De Simone and Olbert, 2021), or if the intended effect of the GMT is diminished in other ways. In such a scenario, there is likely to be no change in ETRs and consequently also no change in the other outcome variables.

- Scenario (b), described in the second row of Table 1, consists of the GMT having a positive impact on MNEs' ETRs, but no further impact on the other outcomes. Such a scenario may arise if MNEs' investment and employment decisions are relatively insensitive to top-up taxation under the GMT, for instance if low-tax outcomes are not associated with significant reductions to returns on marginal investments. This may occur if the substance carveout means that the GMT does not affect tax rates on marginal investments involving tangible assets and additional employment (Schjelderup and Stähler, 2023; Gschossmann et al., 2025), or if the GMT leads to a costly reduction in profit-shifting activities which, at the margin lead to tax savings which are offset by

reduced profit shifting costs (Juranek et al., 2023), leaving overall incentives unchanged.

- Scenario (c) would occur if an increase in ETRs would be associated with a reduction in investment or employment, in line with a model in which the GMT increases the user cost of capital or increases firms' financial constraints.

- Finally, under Scenario (d), firms may attempt to protect low-tax profits by taking advantage of the substance carveout under the GMT which could act as a production subsidy in some circumstances (Schjelderup and Stähler, 2023). In this case, a relatively smaller or null ETR response may be expected but associated with an increase in tangible asset investment and employment.

[Table 1 here]

In addition to testing for implementation effects, the empirical approach also allows for testing of anticipation (post-announcement but pre-implementation) effects. Although the GMT was announced in late 2021, MNEs in scope of the GMT will not have experienced an increase in their ETRs until the implementation of the GMT in 2024. Effects on ETRs or economic activity estimated in earlier years (i.e., 2022-2023) would therefore be anticipation responses. Since this paper's empirical strategy assigns firms to control and treatment groups based on pre-announcement characteristics such anticipation responses can be estimated. A study on stock market reactions to the GMT suggests the relevance of forward-looking anticipation responses at least at the level of financial market participants (Gómez-Cram and Olbert, 2023).

Overall, the direction and magnitude of anticipation responses on ETRs and economic activity is unclear ex-ante. On the one hand, some MNEs may have begun to change their pattern of profit allocation and taxation across jurisdictions before implementation, hence increasing their ETRs, without necessarily altering underlying economic activity. This may be particularly true if such adjustments are costly and subject to delays in implementation. On the other hand, MNEs may have attempted to engage in shifting behaviour over time to minimise top-up taxation under the GMT and increasing ETRs, implying a reduction in ETRs in 2022-2023 and an increase in 2024. However, the ability to engage in such shifting over time is unclear, and may be limited by the requirement that profits and corresponding taxes are closely matched under the GMT. The timing of a potential investment and employment response under each scenario listed in Table 1 is likewise uncertain. It may depend on the precise nature of MNE decision-making and the channel linking ETRs to other MNE outcomes. If investment decisions are made on the basis of an expected increase in the user costs of capital and reduced forward-looking estimates of returns from investment projects,

investment and employment responses may be observed in the post-announcement but pre-implementation period. On the other hand, if investment decisions are impacted because of reduced cash-flow as a result of increased tax liabilities, impacts may only be expected post-implementation.

4 Data and methodology

4.1 Empirical strategy

This paper uses a difference-in-differences approach to identify the effects of the GMT. As specified in the GloBE rules, only MNE groups with consolidated revenues above EUR 750m are in scope of the GMT (see Section 2). The identification strategy exploits this revenue requirement to assign companies to a treatment and a control group. The DiD strategy compares the outcomes of companies more likely to be in scope of the GMT (the treatment group) with those of companies less likely to be in scope of the GMT (the control group) to identify the impact of the GMT. Since only multinationals are in scope of the GMT, the approach restricts attention to companies with at least one foreign subsidiary, in line with the definition of an MNE in the GloBE rules.

MNE groups with revenues above the EUR 750m threshold are assigned to the treatment group while MNEs with revenues below the threshold are assigned to the control group. MNEs are assigned to the two groups based on their average revenues between 2017 and 2019. MNE status is likewise based on having at least one foreign subsidiary over that period. The assignment to treatment and control groups is therefore based on characteristics which may be considered pre-determined from the point of view of the GMT reform. MNEs with revenues very close to the threshold in the 2017-2019 period are excluded to minimise the risk of members of the control group being affected by the GMT, and vice versa. Since inflation will tend to lead companies initially below the revenue threshold to exceed it in later years, a larger exclusion is applied below the cut-off.[5] In the baseline sample used in this paper, the treatment group comprises MNE groups exceeding EUR 800m in revenues, up to a cutoff of EUR 2,000m, as illustrated by Figure 4. The control group consists of companies with revenues between EUR 300m-EUR 650m. The lower and upper cutoffs are defined to ensure companies are comparable while allowing for sufficient sample sizes for reasonable estimation of effects. The exact bounds are varied in a series of robustness checks to ensure these definitions are not driving the results.

[5]The exclusion of companies close to the threshold also helps to avoid potential confounding of treatment-control assignment with unobserved characteristics related to the introduction of Country-by-Country reporting, as discussed below.

[Figure 4 here]

The DiD estimation strategy relies on comparing changes in the outcomes of the two groups over time, while controlling for potential confounding factors. The following model is estimated at the level of MNE i and year t:

$$Y_{it} = \sum_{\tau=2016}^{2024} \theta_\tau \cdot \mathbf{1}[t = \tau] \times \text{Treated}_i + \alpha_i + \mu_t + X'_{it}\delta + \varepsilon_{it} \tag{1}$$

where Treated_i is a dummy equal to one if MNE i is assigned to the treatment group. The coefficients θ_τ measure the change in the outcome variable Y_{it} for treated companies relative to companies in the control group in year τ. The coefficient for the year 2018 is normalised to zero. A number of additional variables control for potential confounding factors. Company fixed effects, indicated by α_i, control for time-invariant differences across companies. Year fixed effects (μ_t) are included to control for time trends or shocks that affect all companies in the sample similarly. The vector X_{it} comprises a set of additional control variables, including industry-year fixed effects to account for industry-specific time trends, interaction terms between dummies for the accounting practice of the company and the different sample years to account for some potentially relevant changes in accounting rules applicable to some companies within the sample period, and GDP growth in the companies' headquarter jurisdiction to control for the impact of the economic cycle.[6] Standard errors in the regressions are clustered at the MNE level.

The regression design can identify both potential announcement effects and implementation effects of the GMT. Announcement effects would be measured through statistically significant differences between the development of the treatment and control group captured via the coefficients θ_{2022} and θ_{2023}.[7] Effects of the implementation of the GMT would be captured by the coefficient θ_{2024}.

The first outcome variable considered is the consolidated ETR. It is calculated as tax expense divided by pre-tax profits at the level of the MNE group. The variable therefore

[6] Even though the companies in the sample are MNEs, the headquarter jurisdiction is likely to play a key role in their global activities. According to aggregated CbCR data published by the OECD, MNEs in scope of the reporting requirement on average report around half their revenues, tangible assets and employees in their headquarter jurisdiction. For smaller MNEs, this share tends to be even higher (OECD, 2024).

[7] Since the GMT was announced later in 2021 (on October 8th and the GloBE Rules being released on 20 December), 2021 is classified as a pre-announcement year, on the basis that most of the year's activities for 2021 for most MNEs would already have been realised at the time of announcement. Nonetheless, the proposed approach allows testing for impacts in 2021 by examining the coefficient in that year (for instance in comparison with the coefficient for 2020).

measures the average tax burden of an MNE, regardless of where profits are reported, or where taxes are paid. ETRs are only calculated for companies with positive profits, to avoid issues in the interpretation of negative ETRs. Observations with negative tax expense are generally included in the sample, but the dependent variables are all winsorised at the 5th and 95th percentile to reduce the impact of outliers. A second set of outcome variables assesses the effect of the GMT on investment and employment growth. The dependent variables used in these DiD estimations are the log-change in tangible fixed assets to measure investment effects, and the log-change in employment to capture effects of the GMT on employment.

A second model is estimated to summarise the average effects of the announcement and implementation of the GMT on treated companies. This second DiD model is specified as follows:

$$Y_{it} = \alpha_i + \mu_t + \theta^A \cdot \text{PostAnnouncement}_t \times \text{Treated}_i + \theta^I \cdot \text{PostImplementation}_t \times \text{Treated}_i + X'_{it}\delta + \varepsilon_{it} \tag{2}$$

where $\text{PostAnnouncement}_t$ and $\text{PostImplementation}_t$ are dummies for the post-announcement period (2022–2023) and the post-implementation period (2024). All other variables are defined as above.

The main identifying assumption behind the DiD estimation strategy is that the outcomes of companies in the treatment and control groups would have followed the same trends in the absence of the introduction of the GMT. The main threat to this strategy is either differential underlying trends between the two groups or time-varying shocks which coincide with the GMT. To validate the strategy, several checks are presented. First, a potential concern is that the introduction of Country-by-Country Reporting (CbCR), introduced starting in 2016, relied on a similar revenue threshold as the GMT.[8] This reform may have led to certain firms choosing to locate below the threshold at least in some years, as documented by Hugger (2024). As discussed above, the GMT analysis in this paper excludes companies with revenues close to the threshold of EUR 750m, where selection around the threshold is most likely. Specifically, the 'donut hole' between EUR 650m and EUR 800m covers the main bunching region identified in Hugger (2024). Additionally, non-random assignment into the treatment and control groups threatens the identification strategy only if it leads to differential underlying trends. The paper therefore tests for parallel trends in the period prior to the announcement of the GMT (see Sections 5.1–5.2). In addition, placebo tests are

[8] Jurisdictions implementing CbCR can opt to use the threshold of EUR 750m or a near equivalent amount in domestic currency (OECD, 2015). However, the reporting requirement for CbCR is based on exceeding the relevant threshold in the previous year and does not rely on the 2 out of 4 criterion applicable for the GMT.

offered, comparing results using the DiD strategy for groups of firms which are either not in scope of the GMT (non-MNEs), for firms which are much less likely to be affected (firms with higher ETRs), and for placebo treatment years (2020 and 2021) (see Section 5.3). All of these tests serve to validate the DiD.

4.2 Data and sample description

The main dataset used for the analysis draws on consolidated financial statements taken from the Orbis database. This data includes all key financials used in the analysis, such as pre-tax profits, tax expense, the number of employees, or the stock of tangible fixed assets and reflects consolidated totals for the MNE groups in the sample. The financial information is paired with historic information on ownership links to construct MNE owner-subsidiary networks on an annual basis. The information on company networks is used to identify multinationals via the existence of foreign subsidiaries, and to identify corporate ultimate owners for each MNE group within a given year. Only observations at ultimate owner level are retained in the dataset since the unit of observation is the full MNE group. The coverage of parent-subsidiary relationships includes links to subsidiaries located in investment hubs and other jurisdictions where unconsolidated financial data is often not available. This information on parent-subsidiary links as well as the consolidated financials used in this paper are generally regarded as high quality (see e.g. Bachas et al. (2023) and Olbert et al. (2024)).[9] Information on the accounting standard used for each financial report and the industry classification of each company are also taken from Orbis. Annual GDP growth rates are compiled from OECD, IMF and World Bank databases.

To ensure that the assignment of MNEs to treatment and control groups accurately captures exposure to the GMT, the sample is restricted in several ways. First, the baseline sample is restricted to non-financial corporate entities, since the revenues listed in the balance sheets of banks and other financial companies do not necessarily align with the revenue definition relevant for the GloBE rules.[10] Second, companies from the shipping sector are excluded, to account for the International Shipping Income exclusion in the GloBE rules. Finally, the GMT applies top-up taxes to companies that have profits taxed at rates below 15% at jurisdiction-level MNE subgroups. Since this paper relies on aggregated data, low-taxed profit cannot be identified at the subgroup level. To overcome this limitation, the baseline

[9] In case individual company networks available in ORBIS are incomplete, this could imply that some MNEs are wrongly identified as non-MNEs. This would reduce the sample size but not systematically bias the results.

[10] In a robustness check, financial companies are included. Although the baseline sample includes some companies that have NACE codes linked to the financial sector, these are mainly holding companies for corporates (NACE code 6420).

sample focuses on companies with below-median consolidated ETRs as such companies are more likely to also have some low-taxed profit in their constituent subgroups compared to companies with higher average ETRs. In an alternative estimate, we also include MNEs with high consolidated ETRs. While these are less likely to be affected by the GMT, these MNEs could still be subject to top-up taxation on operations in individual low-tax jurisdictions since the GMT operates on a jurisdiction-by-jurisdiction basis. Figure A.1 in the Annex shows the distribution of ETRs for the relevant MNEs.

The dataset used in this analysis covers the years 2016-2024. This period includes five years before the announcement of GMT (2016-2020), the year towards the end of which the agreement was reached and the model rules were published (2021), two full years in which the model rules have been public but not implemented (2022-2023) and the first year following the implementation of the GMT in a large number of jurisdictions (2024, see Section 2). Consistent with the entry into force of the GMT in many jurisdictions for MNE fiscal years beginning from 1 January 2024, MNE fiscal years are defined based on their starting date. While the dataset is an unbalanced panel, companies remain in the sample for an average 8.2 years of the possible nine years.

The baseline sample used in the DiD estimation comprises 15,390 MNE-year observations from 2,066 unique MNE groups. Of these, 876 are assigned to the treatment group and 1,190 are assigned to the control group. Since the treatment and control groups are defined over one specific period (2017-2019) the same companies can be consistently tracked over time, to explore e.g. anticipatory effects, and to avoid issues related to selection into (or out of) treatment. However, it comes at the cost of some imprecision in the assignment of companies to the two groups. This could lead to a downward bias in the estimates as some MNEs in the control group are treated, and vice versa. Nonetheless, the assignment predicts very well actual exposure to the GMT. The accuracy of the assignment of MNEs into the two groups is visualised in Panel A of Figure A.2 in the Annex. Even though the sorting is based on 2017-2019 revenues, the probability for MNEs in the treatment group to exceed the revenue threshold of EUR 750m is above or close to 90% in most sample years. For the control group, although this probability increases somewhat over time due to revenue growth and inflation, it is substantially lower. At the same time, the definition of MNE status based on the years 2017-2019 seems to be a very strong predictor for the MNE status across all years of the sample period (Panel B).

By construction, the MNE groups in the treatment group tend to be larger than those in the control group, but the two groups are similar in terms of the key outcomes. Table 2 reports summary statistics for a set of financial variables for the treatment and the control groups separately. Given the threshold-based assignment, MNEs in the treatment group

have higher revenues as well as higher levels of tangible fixed assets and employees than those in the control group. The regression analysis controls for these differences between companies in the two groups by including company-level fixed effects. The key outcome variables considered in this paper are uncorrelated with MNE size and are largely comparable between the two groups. The average ETR, for instance, is 12.9% in the treatment group and 12.4% in the control group, the log changes in tangible assets and employment are also comparable. The correlations between average 2017-2019 revenues and the three outcome variables are all close to zero (correlation coefficients between 0 and 0.05). Finally, the distribution of companies in the treatment and control group across industries (NACE letter codes) is shown in Figure A.3. This distribution is also very similar across the two groups.

[Table 2 here]

5 Results

The section proceeds in steps to analyse the potential announcement and implementation effects of the GMT in line with the methodology outlined in Section 4. Section 5.1 focuses on the effects of the GMT on consolidated ETRs. Any observed changes in these ETRs could in turn be associated with responses in MNEs' total investment and employment, which are estimated in Section 5.2. The robustness of the baseline results reported in Sections 5.1 and 5.2 is tested in Section 5.3 . Section 5.4 studies potential heterogeneities in the responses to the GMT across MNEs with different characteristics and Section 5.5 looks at additional potential margins of response related to the structure of in-scope MNEs.

5.1 Effects of the GMT on consolidated ETRs

Trends in the raw means of the ETRs provide a first indication of the potential effects of the GMT. Panel A of Figure 5 plots the raw means of MNE-level ETRs from 2016 to 2024, where the black solid line represents the mean of MNEs in the treatment group, and the red dashed line represents the mean of MNEs in the control group in each year. The outcome variable is normalised to zero in 2018. Prior to 2021, the trends of ETRs look entirely parallel between the treatment and control groups. Starting in 2021, there is a slight divergence with ETRs in the treatment group trending somewhat above those in the control group. This divergence, however, becomes substantially larger after the implementation of the GMT in the last year of the sample period.

The corresponding DiD estimates test whether these developments are driven by the announcement or implementation of the GMT or whether they can be attributed to confounding factors. The results for this difference-in-differences estimation are presented in Panel B of Figure 5. The figure plots the coefficients $\hat{\theta}_\tau$, where $\tau \in \{2016, \ldots, 2024\}$, from the specification in Equation (1) (see Section 4). The dependent variable is the consolidated ETR. The coefficient estimates are statistically insignificant between 2016 and 2021, supporting the assumption of parallel pre-trends underpinning the estimation strategy. Coefficients also remain insignificant between 2022 and 2023, with point estimates very close to zero, implying no statistically significant announcement response on the consolidated ETRs of in-scope MNEs. In contrast, the coefficient for 2024 is positive and statistically significant at the 1% level, indicating an increase in consolidated ETRs after the implementation of the GMT.

[Figure 5 here]

Table 3 summarises the regression results, with Column (1) showing the coefficient estimates for ETRs following Equation (1). These estimates suggest an increase in consolidated ETRs of around 1.9 ppts post implementation for companies exposed to the GMT compared to companies below the revenue threshold. The DiD estimation based on Equation (2) is reported in Column (4) of Table 3. This specification also suggests no announcement effect (coefficient $\hat{\theta}^A$ not being significantly different from zero, with a point estimate of -0.002), but a statistically significant positive implementation effect with a point estimate of a 1.7 ppts increase in the ETR compared to the pre-implementation period. This point estimate is equivalent to a 12% increase in the consolidated ETR relative to the pre-reform treated mean.

[Table 3 here]

Figure 6 plots DiD estimation results based on Equation (1), where the dependent variables are the log change in tax payments (Panel A), and the log change in pre-tax profits (Panel B). Effectively separating the two components of the ETR helps to disentangle whether the effects of the GMT are driven by changes in the numerator (tax payments) or the denominator (pre-tax profits). The results reported in Figure 6 suggest that the increase in ETRs of in-scope MNEs is caused by a relative increase in their tax expense in 2024 and not a change in pre-tax profits, since profit levels did not decline in the treatment group relative to the control group. The corresponding coefficients are shown in Columns (1) and (2) of Table A.1 in the Annex. The magnitude of the estimated log-change in taxation

observed is also consistent with the observed change in ETRs. [11] The estimation results based on Equation 2, reported columns (4) and (5) of Table A.1, confirm these findings. [12] The increase in ETRs is in line with Scenarios (b) and (c) in Table 1 which both predict an ETR response.[13] Scenarios (b) and (c) differ, however, in their assumptions regarding potential changes in investment and employment growth. These responses are investigated in the following subsection.

[Figure 6 here]

5.2 Effects of the GMT on investment and employment

Changes in the tax burden of companies in scope of the GMT could trigger changes in total investment and employment. The effects of the GMT on the investment and employment growth of in-scope MNEs are again assessed based on the DiD specification presented in Section 4.

The raw means for investment and employment growth are plotted in Panels A and C of Figure 7. These suggest similar developments for the treatment and control groups over the sample period. In line with the perception of a low-growth environment in recent years (see e.g., Dlugosch et al. (2025)), downward trends in investment and employment growth for both groups can be observed. Investment experienced comparably high volatility in the sample period. Part of this was likely driven by adjustments in IFRS accounting rules which changed the accounting for leased equipment and led to an increase in tangible fixed assets reported by affected companies from 2018 to 2019.[14] The DiD estimates presented in Panels B (investment) and D (employment growth) for pre-reform years both confirm that the parallel trend assumption for the pre-GMT period holds. For both measures of MNE activity, the estimations also show no statistically significant response to GMT-announcement or implementation. Columns (2) and (3) of Table 3 summarise the coefficient estimates for the individual years. The coefficients for the specification in Equation 2, showing average effects

[11] The implied percentage change in the total tax expense is 10.8%, which is very close to the 12% increase in the consolidated ETR implied by the baseline estimates.

[12] The observed increase in the tax expense of companies affected by the GMT is consistent with both a direct effect of top-up taxation and indirect effects such as a reduction of corporate profit shifting towards low-tax jurisdictions. Disentangling between the two would require analysis using data at the subgroup level, which is not sufficiently widely available in the dataset used in this paper.

[13] These results are also in line with increasing anecdotal evidence of MNEs increasing tax expenses in response to the GMT in their annual reports.

[14] In the DiD regressions, the effect of such changes in accounting principles is controlled for by the inclusion of an interaction term between dummies for the accounting standards and individual year-dummies, but the changes can lead to greater variation in year-over-year changes of raw means.

of GMT announcement and implementation compared to the pre-GMT period, are reported in columns (4) and (5). Again, they suggest no announcement or implementation effects.[15]

[Figure 7 here]

Overall, the results presented above are in line with Scenario (b) of Table 1 in Section 3 which predicts a positive impact of the GMT on ETRs with no changes in investment or employment. The estimates presented appear to rule out an investment response to the GMT which might have been expected based on recent estimates of the investment effects of unilateral reforms in the literature. Zwick and Mahon (2017), for example, document an elasticity with respect to the net of tax rate of around 4. Ohrn (2018) finds that a 1 percentage point reduction in tax rates implies a 4.7 percent increase in installed capital. Based on these estimates, the increase in ETRs of 1.7 ppts for our baseline sample would imply an 8% reduction in investment.[16] The point estimate reported above of 0.008 (0.002) suggest potential negative changes to investment (employment growth) do not exceed 1.1% (1.4%) based on the 95% confidence interval. This result could be due to the design of the GMT and its substance-based carve-out, which reduces its impact on activities associated with high investment or employment. Alternatively, investment and employment effects could only materialise in the longer run. This may be the case, for instance, if rather than being based on a calculation of post-tax returns on investment, MNEs make investment decisions based on the tightness of financial constraints, which may be only gradually tightened due to the additional tax liability. Such financing constraints, however, might be more relevant for smaller firms than those in-scope of the GMT.

5.3 Robustness and placebo tests

A series of robustness checks support the internal validity of the findings presented above. These exercises test whether the results described are robust to changes in the exact sample used, the set of control variables included, and the definition of the treatment and control group. Results of these robustness checks are summarised in Table 4 for ETRs and Table A.2 in the Annex for investment and employment growth. They generally validate the results from the baseline estimations.

First, the baseline results are robust to several alternative estimation samples. Including

[15] Additionally, as shown in Figure A.4 and columns (3) and (6) of Table A.1, there is no statistically significant announcement or implementation effect of the GMT on investment in intangible assets.

[16] The change in the net of tax rate is around 2% for our baseline estimate, calculated as given the pre-reform treated mean ETR of 14.5% implying a net of tax rate of 85.5%.

companies that did not have below-median ETRs throughout the 2017-2019 period in the estimation sample reduces the point estimate of the ETR effect to 1.4 ppts, compared to 1.7 ppts in the baseline, but the estimate remains statistically significant (columns (1) and (2) of Table 4).[17] The results from this sample also suggest no significant announcement or implementation effect of the GMT on investment or employment. Including banks and other financial companies in the sample results in a slightly lower, but still statistically significant change in ETRs of 1.4 ppts (column (3)), with no statistically significant effects for investment and employment growth.[18] In a further test, companies headquartered in the US are excluded. These MNEs benefited from a temporary Safe Harbour in 2024 reducing the effect of the GMT. In 2026, a Side-by-Side agreement was reached (see Section 2). Top-up taxes on US MNEs are therefore only levied through QDMTTs insofar as they are implemented by the jurisdictions in which affiliates of US MNEs operate. As reported in column (4) of Table 4, the estimated treatment effect of 1.9 ppts on the ETRs for a sample excluding US MNEs is slightly larger than the baseline result. This is in line with the assumption that the GMT only changed the taxation of a portion of profits of US MNEs, who have been subject to a separate system of top-up taxation since 2017, with the exact share depending on the global distribution of activities of each MNE. Using a balanced panel of companies, such that they are observed in each year of the sample period, yields a very similar set of coefficients on the effect of the GMT on ETRs (column (5)). Results for investment and employment effects with the above robustness checks are also similar to the baseline.

Second, the results are robust to the inclusion of a series of additional control variables to the baseline model. The inflation rate of the headquarter jurisdiction is used to further proxy for economic uncertainty faced by the MNEs. This variable can also help to mitigate potential effects of exchange rate changes on the euro-denominated outcome variables. To control more directly for potential currency effects, the interaction between a set of dummies for the original reporting currency of the MNE and year dummies is included. Different paths in terms of total profitability or the development of revenues between the treatment and control group could also lead to differences in outcomes over time. To alleviate such concerns, the return on revenues or the log of lagged revenues are tested as additional MNE-level controls. In a further test, headquarter times year fixed effects are included as alternative means to control for headquarter-specific time trends. None of these additional controls lead to substantial changes in the results compared to the baseline estimates (see

[17] In Section 6, this coefficient is used to estimate the revenue impact of the GMT.

[18] The lower coefficient could be related to the issues around imprecision in the assignment to treatment and control for financial companies, as discussed in Section 4.2

columns (6) to (10) of Table 4 and Table A.2).

Third, the results are robust to alternative revenue bounds used to define the treatment and control groups. The DiD approach compares MNEs below the threshold with MNEs above the threshold. Raising the minimum revenue for the control group from EUR 300m to EUR 400m and reducing the maximum revenue for the treatment group from EUR 2,000m to EUR 1,000m makes the two groups more comparable in terms of size. However, it comes at the cost of reducing group sizes and therefore the statistical power of the estimates. Nevertheless, these adjustments (or each of them individually) do not lead to any relevant changes compared to the baseline results (column (11) of Table 4 and Table A.2). As shown in column (12), entirely removing the upper bound also results in a similar estimate on the treatment effect on ETRs.[19] Extending the "donut hole" to exclude companies between EUR 500m and 900m (instead of EUR 650m-800m) further reduces the probability of companies in the control group being treated, and vice versa. It does, however, not substantially change any of the baseline results (column (13)). In addition to the revenue bounds defined in the 2017-2019 period, the assignment into the treatment and control group can also be based on the two-out-of-four criterion using revenue in the years 2020-2023. Since these revenues are not pre-determined at the time of the announcement of the GMT, they could be impacted by MNE responses to the GMT. At the same time, this criterion more closely aligns with the GloBE rules. Column (14) shows that the resulting estimates on the impacts of the GMT are still very comparable to the baseline results.

In addition, the data are used to construct a series of placebo tests to validate the baseline results. For this, the estimations are first repeated with companies that have no foreign subsidiary listed in Orbis. Since these companies are not MNEs according to the definition of the GloBE rules, they are not in scope of the GMT. They should be relatively unaffected by its introduction, even if they are above the revenue threshold.[20] In line with this expectation, the corresponding DiD estimates reported in column (2) of Table 5 show no statistically significant treatment effect on the ETRs of non-MNEs above the revenue threshold compared to non-MNEs below the threshold. Results for investment and employment growth are reported in Table A.3 in the Annex and also yield insignificant treatment effects. As a second placebo test, the estimations are repeated with a sample containing only previously excluded MNEs that did not have below-median ETRs throughout the 2017-2019 period. These MNEs could be in scope of the GMT depending on their revenue levels but may be less likely to be subject to larger amounts of top-up taxation. For these companies, the

[19]Section 5.4 further explores potential heterogeneities in the treatment effect across the revenue distribution.

[20]Non-MNEs headquartered in the EU or the UK are also excluded from this exercise since these headquarter jurisdictions also apply the GMT on large domestic groups (see Council Directive (EU) 2022/2523).

treatment effect on ETRs is only borderline significant with a point estimate that is around a third lower than for the baseline sample (column (3)). Estimated treatment effects for investment and employment growth are again not statistically significant (Table A.3). This is in line with the assumption that MNEs with higher consolidated ETRs are less affected by the introduction of the GMT. Finally, investigating the coefficients for pre-reform years (e.g. 2020 or 2021) from the baseline estimations can be interpreted as placebo tests assuming earlier reform years. For these coefficients, the estimated treatment effects on the main outcomes are all not statistically significantly different from zero (see columns (1)-(3) of Table 3).

[Table 4 here]

[Table 5 here]

5.4 Heterogeneity

To assess whether the effects documented above vary across different types of firms, the analysis explores heterogeneity along several sectoral and firm-level dimensions. Differences are examined by sectoral indicators of potential tax planning aggressiveness and by a measure of profit-shifting capacity. Heterogeneity is also assessed by the ratio of profits to tangible fixed assets, as a proxy for profits reported relative to economic substance, by firm size, and by the GMT implementation status of the headquarter jurisdiction.

A heterogeneity analysis of ETR responses based on splitting the sample based on MNE presence in sectors with relatively higher or lower levels of aggressive tax planning intensity is presented in columns (2) and (3) of Table 6. The classification of MNEs is based on previous findings about which sectors display the most-aggressive tax practices (Altshuler et al., 2024).[21] In-scope firms in sectors with greater propensity for higher tax aggressiveness show an increase of 2.9 ppts in their consolidated ETR following the implementation of the GMT, whereas the effect for firms in a sector with a lower propensity to be tax aggressive is statistically insignificant. Moreover, heterogeneity is also examined by firms' intangible asset intensity. Prior research has identified intangible-intensive firms as having a greater propensity for profit shifting, as intangible assets are relatively mobile and often lack ob-

[21] Altshuler et al. (2024) identify the probability for MNEs to adopt hybrid tax planning strategies by sector. For the sample split, MNEs are assumed to have a high probability for aggressive tax planning if they are active in a sector in which hybrid tax planners are over-represented. According to Altshuler et al. (2024) such sectors are Data Processing, Hosting, and Related Services; Other Information Services; Miscellaneous Manufacturing; Publishing Industries (except Internet); Merchant Wholesalers, Nondurable Goods; Chemical Manufacturing; Computer and Electronic Product Manufacturing; Primary Metal Manufacturing.

servable market prices, which can ease the allocation of profits across jurisdictions (Delis et al., 2021). Columns (4) and (5) show that in-scope firms in the top quartile of the distribution of the ratio of intangible assets to total assets experience a statistically significant increase of 2.3 ppts in their consolidated ETR following the GMT. However, no statistically significant effects are observed for firms in the bottom quartile. The impact of the GMT on ETRs therefore appears strongest among firms in sectors previously identified as more likely to engage in aggressive tax planning practices and to have greater profit-shifting capacity. Finally, there does not appear to be a significant heterogeneity in impacts on employment and investment, neither by sector nor by intangible asset intensity, as reported in Table A.4 in the Annex.

[Table 6 here]

Impacts on ETRs also seem to be larger for firms which report more profits relative to an estimated amount of substance. Estimates in columns (6) and (7) of Table 6 show that in-scope firms in the top quartile of the profits-to-tangible-fixed-assets distribution, a proxy for profits relative to economic substance, experience an increase in consolidated ETRs of around 2.9 ppts following implementation, while the corresponding estimate for those in the bottom quartile is not statistically significant. This pattern is consistent with the design of the GMT which targets profits in excess of substance within the top-up tax calculation. As shown in Table A.4 in the Annex, there are no significant differential effects on investment or employment between firms with higher and lower profit-to-tangibles ratio.

[Table 7 here]

There does not appear to be differential GMT impacts on ETRs by size within the baseline sample. Within the baseline sample, as shown in column (1) of Table 7, MNEs below and above the median revenue of the treatment group exhibit estimated implementation effects of 1.69 and 1.80 ppts, respectively, with no statistically significant difference between the two. Similarly, columns (1) and (3) of Table A.5 in the Annex suggest no impacts on investment or employment for both larger and smaller MNEs in the treatment group. Column (2) of Table 7 reports results for a sample where MNEs with revenues above EUR 2,000m are included in the sample. The results indicate that larger firms also experience a positive effect on consolidated ETRs of similar size.[22] Additionally, columns (2) and (5) of Table A.5 report estimated positive impacts on investment and employment growth for the largest firms,

[22]The parallel trend assumption also holds for these larger MNEs for all three main variables of interest.

suggestive of potential responses by MNEs seeking to take advantage of the substance-based carve-out under the GMT, thereby protecting low-taxed profits from top-up taxation. This result would be in line with Scenario (d) in Table 1 in Section 3. The results for the largest companies, however, have to be treated with more caution since they rely on comparing control-group companies with companies further away from the revenue threshold reducing the comparability between the two groups.[23]

Most companies not in scope of an IIR are only subject to top-up taxation through QDMTTs in 2024.[24] This might result in only a fraction of total profit being covered by the GMT, depending on the exact allocation of profits within the individual MNE.[25] Column (3) of Table 7 presents the estimated coefficients when the treatment group is split into MNEs subject to an IIR and MNEs from headquarter jurisdiction that have not introduced an IIR in 2024. For MNEs subject to an IIR, the estimated treatment effect on the consolidated ETR is 2.9 ppts, so around two-thirds larger than for the full sample. In contrast, MNEs not subject to an IIR show a weaker ETR response, with a point estimate of 1.3 ppts which is only statistically significant at the 10% level.

5.5 Additional margins of response

The GMT might not only affect financial outcomes, but also the structure of in-scope MNEs. MNEs in scope of the GMT could, for instance, reduce the number of their subsidiaries to lower organisational complexity and potentially compliance costs. Reduced profit shifting incentives due to top-up taxation could also lead MNEs to reduce their presence in investment hubs.

The analysis conducted in this paper does not find effects of the GMT on MNE structures. Figure 8 summarises the coefficient estimates for outcome variables related to MNE structure, namely the total number of subsidiaries of MNE groups (Panel A) and the number of subsidiaries located in investment hubs (Panel B) following Equation 1. For both measures, the coefficient estimates suggest no statistically significant announcement or implementation effect. This is confirmed by additional regression estimates based on Equation 2 which are reported in Table A.1. The results suggest that in-scope MNEs have not responded to the GMT with a strong restructuring of their subsidiary structure before or immediately after

[23]Despite the increased difference between the revenues of the MNEs in the control group and the MNEs above the EUR 2,000m threshold, the parallel trend assumption also holds for these larger companies across all three main outcome variables (ETR, investment, and employment).

[24]EU member states that have implemented a UTPR have levied top-up taxes through the UTPR on other EU members that have delayed implementation of the GMT.

[25]For the year 2024, the UTPR did not apply due to deferred implementation with the exception of being levied within the EU.

the implementation of the reform. This null result on MNE structures may suggest that these types of adjustments might take time.

[Figure 8 here]

6 Estimation of increased tax revenue

The estimated ETRs effects imply potentially substantial revenue consequences. Our point estimates can be used to provide first ex-post estimates of additional global tax revenue raised due to the GMT. Table 8 provides a summary of the implied revenue impacts for the year 2024 based on estimated ETR impacts post-implementation. The calculations are based on specification (2) of Table 4 estimated on the population of all MNEs (not just those with low ETRs) with 2017-2019 revenues of EUR 300m-650m in the control group, and EUR 800m-2bn in the treatment group, as in the baseline specification. This approach allows for the consideration of impacts on all MNEs (not just low-taxed ones), while relying on an identification strategy which focuses on MNEs closest to the eligibility threshold.[26] To obtain global tax revenue estimates for 2024, the estimated ETR impact is multiplied by total profits reported by MNEs in various revenue ranges for the year 2024. In column (1) of Table 8, total profits are calculated only for MNEs within the revenue bounds of the treatment group in the estimation sample (EUR 800m-2bn). In columns (2) and (3), total profits are calculated for all MNEs with more than EUR 750m of revenue (i.e. the entire treated population). In column (2), the estimates of total profit are based on Orbis data from 2024, while in column (3) the estimates are based on aggregated CbCR data, complemented with additional sources.[27]

Using an estimate of total profits for all MNEs with revenues above EUR 750m, the point estimates suggest global tax revenue could have increased in 2024 by EUR 79-109bn, with 95% confidence intervals of EUR 22-187bn. The estimates based on total profits restricted to the baseline estimation sample revenue range are the lowest, at EUR 4bn with a EUR 1-7bn 95% confidence interval. Once the estimated ETR impacts are applied to all MNEs

[26] The strategy involves extrapolating ETR impacts from MNEs with revenues below EUR 2bn in 2017-2019 to those with revenues above EUR 2bn. The statistically insignificant differences in responses for the largest MNEs compared with the MNEs in the baseline treatment group reported in Table 7 lends support to this strategy.

[27] The combination of aggregated CbCR data with additional sources such as FDI statistics follows Hugger et al. (2023) with additional adjustments to exclude loss-making MNEs. Since aggregated CbCR data is only available up to the fiscal year 2022 at the time of writing, CbCR data for 2024 is imputed using changes in total annual profits reported in the consolidated Orbis sample used for this paper.

above EUR 750m in revenue in the Orbis data, the estimated increase in tax revenues is EUR 79bn with a EUR 22-136bn 95% confidence interval. Finally, this estimate is applied to the somewhat higher estimate of total in-scope profits based on aggregated CbCRs which is likely to be most closely aligned with the GMT tax base. Using this data source, a tax revenue impact of EUR 109bn is estimated with a EUR 31-187bn 95% confidence interval. These estimates represent an increase of between 2.4% and 3.4% of total global CIT revenue, relative to 2023 values. [28]

[Table 8 here]

Overall, the revenue estimates are somewhat lower than some of the ex-ante estimates, such as those in Hugger et al. (2024), while being of similar order of magnitude. [29] In interpreting results, it should be noted that the ETR impacts estimated in this paper are for 2024 only and are thus unlikely to correspond to the long-term impacts. The estimates reflect the initial SBIE carveout and may rise as this carveout is gradually reduced over time. As discussed above, indirect effects through reduced profit shifting might also take time to materialise if MNEs only gradually adjust their structures. A further factor that could impact the results is the gradual implementation of the GMT. While implementation is already widespread in 2024, many countries are still implementing parts of the GMT and the UTPR will only become effective in 2025 for most MNEs. This means that low-taxed profit in jurisdictions without QDMTT for MNEs not headquartered in an IIR-jurisdiction would be largely exempt from top-up taxation in 2024, reducing revenue below that modelled in Hugger et al. (2024). Furthermore, the 2024 estimates presented in this paper pre-date the Side-by-Side agreement of 2026 which allows for a more generous treatment of certain tax incentives in the calculation of the GloBE ETR. This could result in lower tax revenues through the GMT in later years. There may be additional changes in impact over time, for instance as uncertainty over implementation by various jurisdictions and the final details of the rules is resolved.

7 Conclusion

This paper exploits the introduction of the Global Minimum Tax to provide evidence on how MNEs respond to coordinated corporate tax reform. The GMT's design and internationally

[28] Total global CIT revenue for 2023 is EUR 3,250bn based on OECD and IMF data.

[29] The estimates for the first post-implementation year are USD 155-192bn in Hugger et al. (2024), in comparison to the USD 86-118bn range of point estimates (95% confidence interval of USD 24-203bn) based on columns (2) and (3) in Table 8.

coordinated implementation create a unique setting for such an analysis. The results therefore complement and extend the literature estimating the real effects of corporate taxation from country-specific policy changes. By utilising data covering the post-announcement and post-implementation period, this analysis also moves beyond the previous literature on the GMT itself which mainly consists of ex-ante assessments. It also represents a first empirical assessment of the impacts of a comprehensive corporate tax reform coordinated across a wide set of jurisdictions.

The results indicate increased ETRs of in-scope MNEs post implementation, but this increase was not accompanied by detectable reductions in investment or employment. The increase in consolidated ETRs of in-scope MNEs after the implementation of the GMT in 2024 was driven primarily by an increase in tax liability. MNEs likely to be most affected by the reform given relatively low pre-GMT ETRs experienced an increase in their consolidated ETRs of 1.7 ppts. The increase for all firms is 1.4 ppts. These effects could either be driven through the direct effect of top-up taxation, or by a reduction in tax-motivated shifting of profits towards low-tax jurisdictions. The estimates are also used to approximate the change in corporate tax revenues due to the GMT, suggesting an additional EUR 79bn-109bn of CIT revenue globally in 2024. This represents an increase in global CIT revenue of 2.4-3.4%. However, the findings on economic activity suggest that the GMT did not induce a negative investment or employment response by MNEs, at least based on data from the first post-implementation year. The paper further finds that the changes in ETRs are more concentrated among MNEs in sectors previously found more likely to contain firms engaging in profit shifting activity, among MNEs with more profit shifting potential, as well as MNEs with higher levels of profit relative to substance. Nonetheless, there are also no significant effects on the count of subsidiaries or the presence in investment hubs for in-scope MNEs. More broadly, the results suggest that coordinated international tax reform can increase effective taxation of MNEs while avoiding short-run reductions in real economic activity that might be expected from evidence based on unilateral tax increases.

The analysis conducted can help to inform future international tax policy and continuing analysis of the GMT. At the same time and given the limited data available on the post-implementation period, the results represent short-run effects. MNEs might adjust their behaviour only gradually over time. The reduction of the substance exclusion over the coming years set out in the GloBE rules and the full entry into force of the UTPR might also induce the effects of the GMT to become stronger over time as the share of MNE profits under the GMT rises further. The implementation of the Side-by-Side agreement of 2026 and other recent changes could also impact the effects of the GMT in some jurisdictions.

Future analysis based on additional years of data will be necessary to complement the

findings presented in this paper with a longer-run perspective. Additional research could also clarify the role of the substance carve-out which implies no change in the taxation of profits stemming from substance-heavy activities. It is also possible that while MNEs did not respond in terms of total investment, they have responded with a reshuffling of activities across jurisdictions. Analysis using jurisdiction-level data will be important to determine the extent of such responses. As additional post-reform data becomes available, the GMT will continue to provide a valuable setting for understanding how MNEs respond to coordinated reforms in an integrated global economy.

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Tables

Table 1: Possible effects of the GMT

	ETR	Investment (fixed assets)	Employment
a. MNEs ETRs not impacted	No impact	No impact	No impact
b. MNEs insensitive to the GMT	(+)	No impact	No impact
c. MNEs sensitive to the GMT	(+)	(−)	(−)
d. MNEs increase substance to protect low ETRs	(+)/No impact	(+)	(+)

Note: The table considers three key scenarios of impacts of the GMT on MNE ETRs and economic activity. (+) denotes an anticipated increase in the outcome variable while (−) denotes an anticipated decrease in the outcome variable.

Table 2: Descriptive statistics

	Treatment group	Control group
Revenue (EUR, m)	1271.6	441.9
Profit before tax (EUR, m)	136.2	46.0
Tax expense (EUR, m)	17.2	6.1
Tangible fixed assets (EUR, m)	796.8	262.9
Intangible fixed assets (EUR, m)	307.7	78.7
Number of employees	4,254.3	2,612.7
ETR (%)	12.9	12.4
Investment [log change (tangible fixed assets)]	0.113	0.103
Employment growth [log change (number of employees)]	0.0452	0.0501
No. of companies	876	1,190
No. of company-year observations	6,629	8,761

Note: Financial data shows averages by group for the years 2017–2019. The outcome variables (ETR, investment and employment growth) are winsorised at the 5th and 95th percentile.

Table 3: Baseline regression results

	(1) ETR	(2) Investment	(3) Employment	(4) ETR	(5) Investment	(6) Employment
Treated × 2020	0.00251	0.00800	0.0114			
	(0.00603)	(0.0114)	(0.0104)			
Treated × 2021	0.00420	0.00751	0.00294			
	(0.00541)	(0.0119)	(0.0100)			
Treated × 2022	−0.00163	0.0128	−0.000852			
	(0.00591)	(0.0114)	(0.0102)			
Treated × 2023	−0.000119	0.00961	0.00855			
	(0.00681)	(0.0114)	(0.00986)			
Treated × 2024	0.0187**	0.0146	0.00452			
	(0.00772)	(0.0122)	(0.0107)			
Treated × post-Announcement ($\hat{\theta}^A$)				−0.00220	0.00487	0.00170
				(0.00439)	(0.00758)	(0.00616)
Treated × post-Implementation ($\hat{\theta}^I$)				0.0174**	0.00832	0.00243
				(0.00695)	(0.00988)	(0.00830)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year and company fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Pre-reform treated mean	0.145	0.0851	0.0420	0.145	0.0845	0.0420
Observations (firm-years)	12,324	14,592	10,940	12,324	14,592	10,940

Note: Columns (1) to (3) report estimated coefficients $\hat{\theta}_\tau$ where $\tau \in \{2016, \ldots, 2024\}$ from Equation (1). Columns (4) to (6) report the estimated coefficients $\hat{\theta}^A$ and $\hat{\theta}^I$ from Equation (2). The dummy variable Treated_i takes the value of 1 for companies expected to be in scope of the GMT. All specifications include year and company fixed effects as well as additional controls following Equations (1) and (2). Standard errors are clustered at the company level and are reported in parenthesis.

Table 4: Robustness checks – ETRs

	(1) Baseline	(2) Incl. higher taxed MNEs	(3) Incl. financial companies	(4) Excl. US MNEs	(5) Balanced sample	(6) Add. control: inflation	(7) Add. control: currency	(8) Add. control: profitability	(9) Add. control: lagged revenue	(10) Add. control: HQ x year	(11) Narrow revenue bounds	(12) No upper revenue bound	(13) Alternative donut hole	(14) Two out of four criterion
Treated × post-Announcement	−0.00220	−0.00350	0.000653	−0.00265	−0.00228	−0.00227	−0.00190	−0.00252	−0.00230	−0.00214	0.00115	−0.00425	−0.00467	−0.00478
	(0.00439)	(0.00328)	(0.00408)	(0.00429)	(0.00525)	(0.00465)	(0.00436)	(0.00435)	(0.00440)	(0.00381)	(0.00698)	(0.00383)	(0.00500)	(0.00421)
Treated × post-Implementation	0.0174**	0.0136***	0.0135**	0.0188***	0.0194**	0.0170**	0.0152**	0.0178**	0.0178**	0.0154***	0.0213**	0.0184***	0.0178**	0.0213***
	(0.00695)	(0.00497)	(0.00633)	(0.00683)	(0.00817)	(0.00745)	(0.00701)	(0.00693)	(0.00696)	(0.00539)	(0.00938)	(0.00616)	(0.00785)	(0.00703)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year and company FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations (firm-years)	12,324	37,791	13,544	11,409	8,557	11,525	12,324	12,301	12,869	12,299	5,584	18,672	9,483	15,581

Note: This table presents robustness checks for the estimates of ETR impacts. The different columns report estimated coefficients $\hat{\theta}^A$ and $\hat{\theta}^I$ from Equation (2) for a number of alternative samples. Some estimations include additional or alternative control variables as indicated in the main text (columns (6)–(10)). The dummy variable Treated_i takes the value of 1 for companies expected to be in scope of the GMT. The dependent variable is the consolidated ETR at the MNE group level. All specifications include year and company fixed effects as well as additional controls following Equation (2). Standard errors are clustered at the company level and are reported in parenthesis.

Table 5: Placebo tests – ETR

	(1) Baseline	(2) Non-MNEs	(3) Non-low ETR
Treated × post-Announcement	−0.00220	−0.0132	−0.00119
	(0.00439)	(0.00935)	(0.00446)
Treated × post-Implementation	0.0174**	−0.0148	0.0124*
	(0.00695)	(0.0112)	(0.00665)
Controls	Yes	Yes	Yes
Year and company fixed effects	Yes	Yes	Yes
Observations (firm-years)	12,324	4,712	25,467

Note: Columns (1) to (3) report the estimated coefficients $\hat{\theta}^A$ and $\hat{\theta}^I$ from Equation (2). The dummy variable Treated_i takes the value of 1 for companies expected to be in scope of the GMT. The dependent variable is the consolidated ETR at the MNE group level. All specifications include year and company fixed effects as well as additional controls following Equation (2). Standard errors are clustered at the company level and are reported in parenthesis.

Table 6: Heterogeneity estimates – ETRs

	(1)	Industry by profit-shifting concentration		Intangible fixed assets over total assets		Profit over tangible fixed assets	
	Baseline	(2) Low	(3) High	(4) Bottom quartile	(5) Top quartile	(6) Bottom quartile	(7) Top quartile
Treated × post-Announcement	−0.00220	−0.00442	0.00606	−0.00529	0.00345	−0.00252	0.00690
	(0.00439)	(0.00533)	(0.00802)	(0.00973)	(0.00941)	(0.0109)	(0.00879)
Treated × post-Implementation	0.0174**	0.0125	0.0286**	0.0143	0.0231*	0.0169	0.0290**
	(0.00695)	(0.00835)	(0.0127)	(0.0160)	(0.0132)	(0.0159)	(0.0144)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year and company fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pre-reform treated mean	0.145	0.148	0.146	0.140	0.160	0.125	0.144
Observations (firm-years)	12,324	8,527	3,619	3,030	3,045	3,095	3,000

Note: The table reports estimated coefficients $\hat{\theta}^A$ and $\hat{\theta}^I$ from Equation (2) for different subsamples. The dummy variable Treated_i takes the value of 1 for companies expected to be in scope of the GMT. The dependent variable is the consolidated ETR at the MNE group level. All specifications include year and company fixed effects as well as additional controls following Equation (2). Standard errors are clustered at the company level and are reported in parenthesis.

Table 7: Heterogeneity estimates – ETRs impact decomposition by size and IIR status

	(1) Split of treatment group by size	(2) Split of treatment group by size & no upper bound		(3) Split of treatment group by IIR implementation
Treated × post-Announcement				
Below median treated baseline sample	−0.00041	0.000313	IIR headquarter	0.0134**
	(0.00558)	(0.00557)		(0.00666)
Above median treated baseline sample	−0.00420	−0.00212	Non-IIR headquarter	−0.0112**
	(0.00535)	(0.00527)		(0.00489)
Above EUR 2,000m		−0.00739*		
		(0.00445)		
Treated × post-Implementation				
Below median treated baseline sample	0.0169**	0.0183**	IIR headquarter	0.0286***
	(0.00832)	(0.00826)		(0.00994)
Above median treated baseline sample	0.0180**	0.0221**	Non-IIR headquarter	0.0132*
	(0.00870)	(0.00856)		(0.00809)
Above EUR 2,000m		0.0179***		
		(0.00680)		
Controls	Yes	Yes		Yes
Year and company fixed effects	Yes	Yes		Yes
Pre-reform treated mean	0.145	0.149		0.145
Observations (firm-years)	12,324	18,960		12,324

Note: The different columns report estimated coefficients $\hat{\theta}^A$ and $\hat{\theta}^I$ from Equation (2). Column (1) reports coefficients separately for firms split by size relative to the median within the treated group. Column (2) extends the sample by including firms with revenues above EUR 2,000m and reports separate effects for this group. Column (3) reports coefficients separately for firms from headquarter jurisdictions with and without IIR. The dummy variable Treated_i takes the value of 1 for companies expected to be in scope of the GMT. All specifications include year and company fixed effects as well as additional controls following Equation (2). Standard errors are clustered at the company level and are reported in parenthesis.

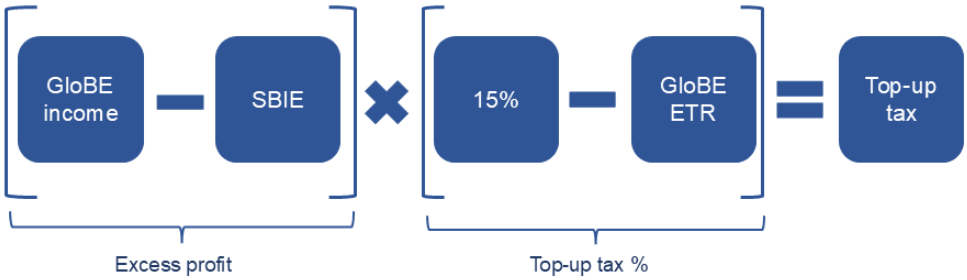
Table 8: Estimates of global tax revenue impacts of the GMT

	(1)	(2)	(3)
1. Estimated ETR impact – point estimate	0.0136***	0.0136***	0.0136***
	(0.00497)	(0.00497)	(0.00497)
2. Estimated ETR impact – 95% CI	0.0038–0.0233	0.0038–0.0233	0.0038–0.0233
3. Total amount of profit in 2024	EUR 298bn	EUR 5,824bn	EUR 8,031bn
	(USD 323bn)	(USD 6,303bn)	(USD 8,692bn)
4. Estimated tax revenue impact (1. × 3.)	EUR 4bn	EUR 79bn	EUR 109bn
	(USD 4bn)	(USD 86bn)	(USD 118bn)
5. Estimated tax revenue impact 95% CI (2. × 3.)	EUR 1–7bn	EUR 22–136bn	EUR 31–187bn
	(USD 1–8bn)	(USD 24–147bn)	(USD 33–203bn)
6. Revenue interval for tax calculations	EUR 800m–2bn	EUR >750m	EUR >750m
7. Source of data for in-scope profit	Orbis (2024)	Orbis (2024)	CbCR (2022, extrapolated to 2024)

Note: The table shows estimated global tax revenue impacts of the GMT for the year 2024 based on the estimated ETR impacts. The ETR estimates used in the calculations are reported in row 1 (point estimates) and row 2 (the 95% confidence interval). All results are based on specification (2) of Table 4 estimated on all MNEs (not just MNEs with low ETRs). The revenue impacts are estimated by multiplying the estimated ETR impact (rows 1 and 2) by the total amount of profit reported by MNEs in 2024 (row 3). Row 4 reports the resulting point estimate of global tax revenue impacts while row 5 reports the implied 95% confidence interval. Column (3) uses 2022 CbCR data, complemented with other sources, to calculate total in-scope profit. This profit amount is adjusted to 2024 values using growth rates observed in Orbis data between 2022 and 2024. Row 6 shows the revenue range for MNEs used to estimate profits in row 3, and row 7 shows the dataset used. Only MNEs with positive profits are included in calculating total profit amounts.

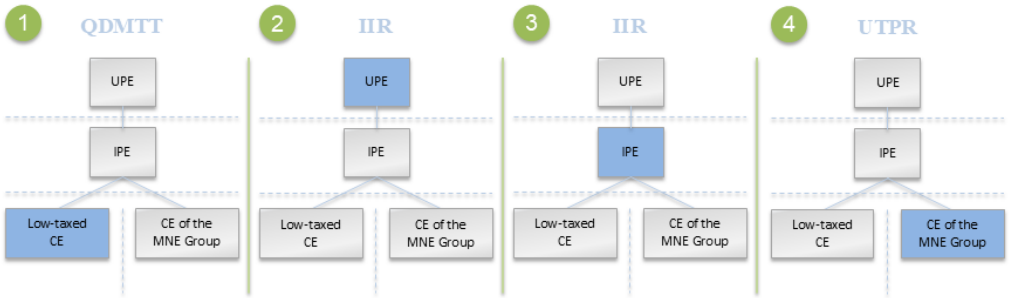
Figures

Figure 1: Top-up tax calculation



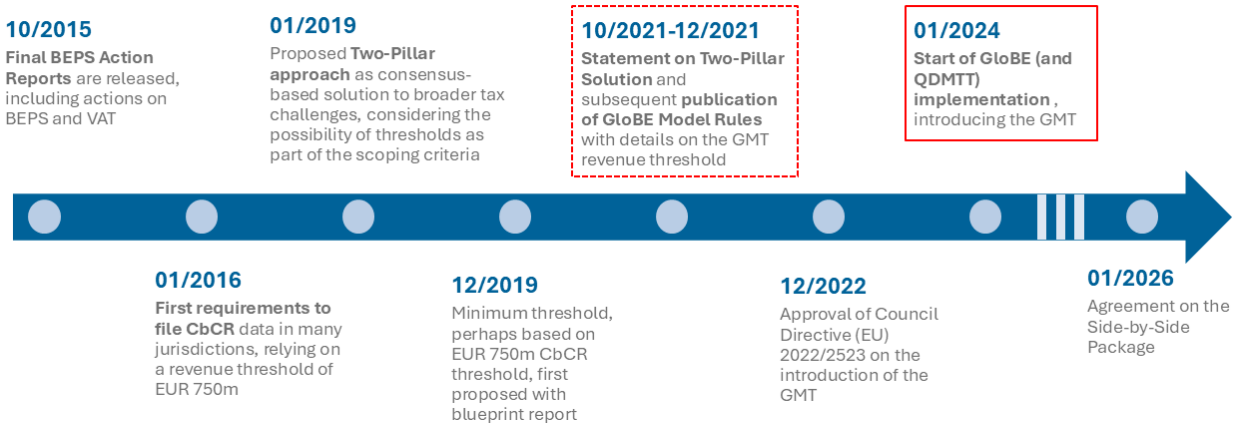
Note: Simplified formula for the calculation of top-up taxes at the subgroup level. *Source:* Hugger et al. (2024).

Figure 2: GMT rule order



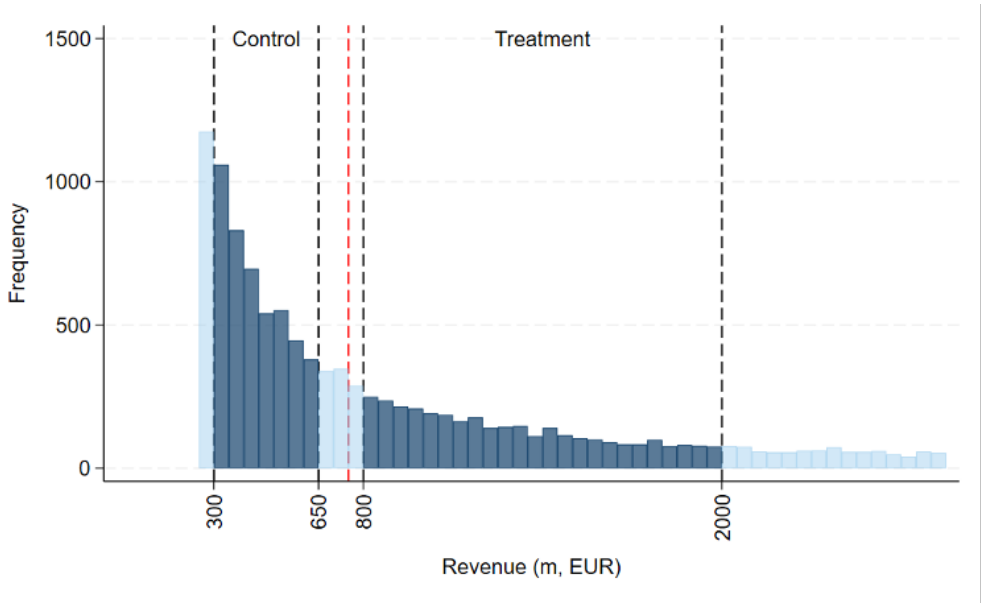
Note: QDMTT: Qualified Domestic Minimum Top-Up Tax. IIR: Income Inclusion Rule. UPE: Ultimate Parent Entity. IPE: Intermediate Parent Entity. CE: Constituent Entity. *Source:* (OECD, 2023a).

Figure 3: Timeline of GMT implementation



Note: High-level summary of selected milestones towards the implementation of the Global Minimum Tax.

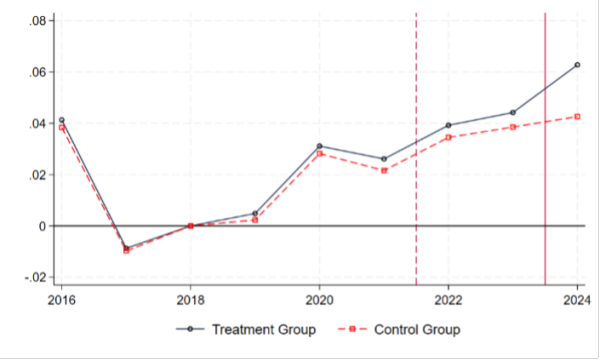
Figure 4: Definition of the treatment and control group across the revenue distribution



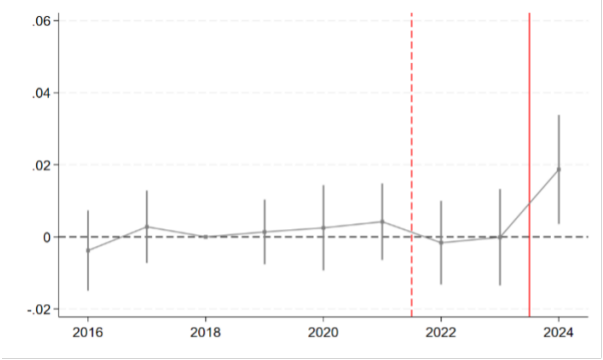
Note: Illustration of the definition of the treatment and control group. The underlying revenue distribution is based on fiscal years 2017–2019 for MNEs from which the two groups are selected.

Figure 5: Raw means and treatment effect for ETRs

Panel A: Raw means



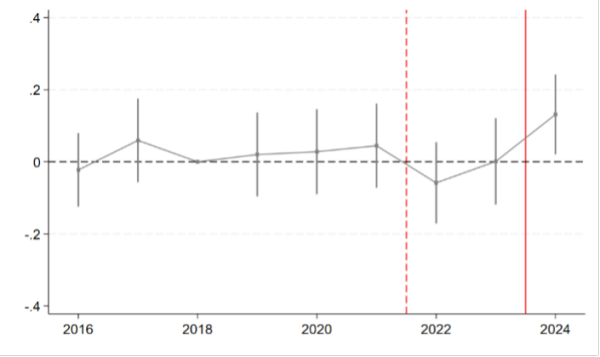
Panel B: DiD coefficients



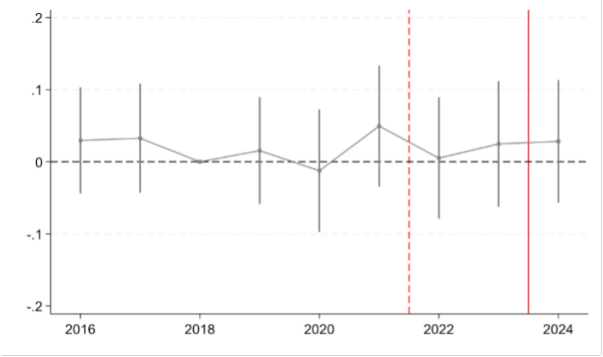
Note: Panel A shows raw means of ETRs for the treatment group and control group in each year of the sample period. The definition of the treatment and control group is based on the EUR 750m revenue threshold (see Section4). Panel B shows the coefficients $\hat{\theta}_\tau$ on $\text{Treated}_i \times \text{Year}_t$ for ETRs, as defined in Equation (1). The vertical lines indicate 95 percent confidence intervals for the coefficient estimates. The dashed red lines in both panels indicate the announcement of the GMT in 2021, the solid red lines indicate the implementation of the GMT in 2024. Outcome variables are normalised to zero in 2018.

Figure 6: Treatment effects for taxation and pre-tax profit

Panel A: Log change taxation



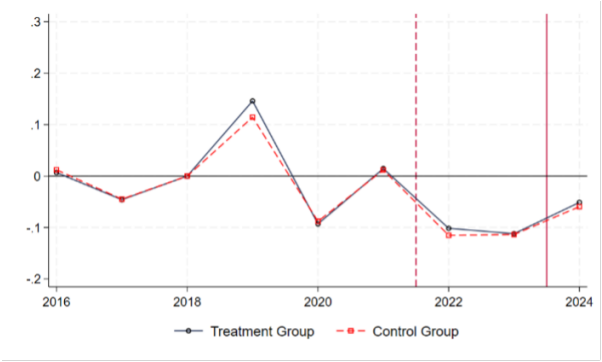
Panel B: Log change pre-tax profits



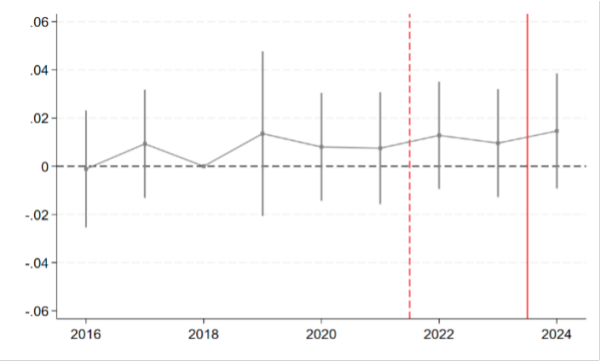
Note: The charts show the coefficients $\hat{\theta}_\tau$ on $\text{Treated}_i \times \text{Year}_t$ as defined in Equation (1) for the log change in taxation (Panel A) and the log change in pre-tax profits (Panel B). The vertical lines indicate 95 percent confidence intervals for the coefficient estimates. The dashed red lines in both panels indicate the announcement of the GMT in 2021, the solid red lines indicate the implementation of the GMT in 2024. Outcome variables are normalised to zero in 2018. The sample only includes observations which are also used for the regressions estimating ETR impacts.

Figure 7: Raw means and treatment effect for investment and employment

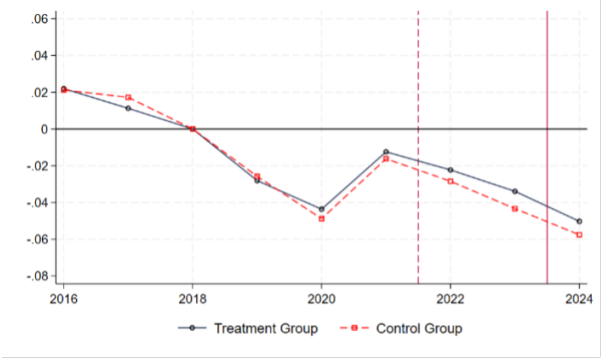
Panel A: Raw means, investment



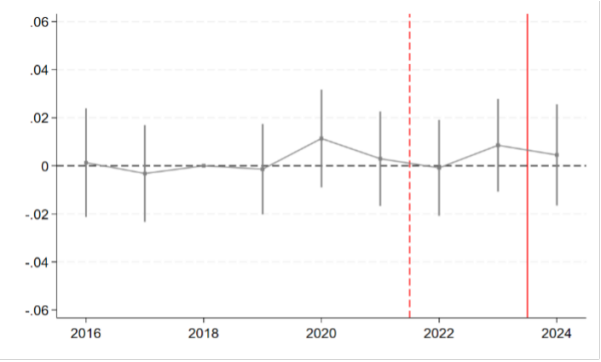
Panel B: DiD coefficients, investment



Panel C: Raw means, employment growth



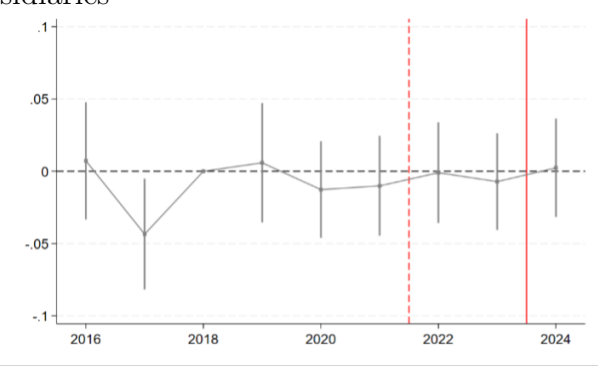
Panel D: DiD coefficients, employment growth



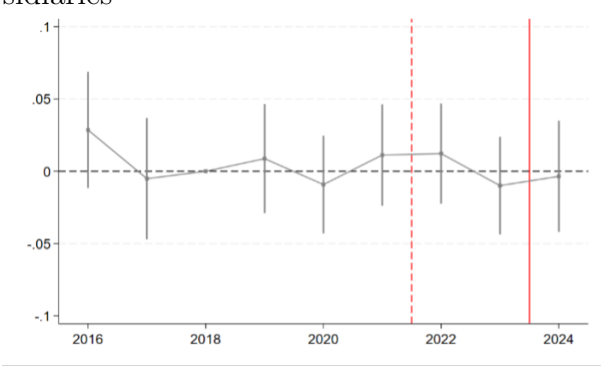
Note: Panels A and C show raw means of log changes for tangible fixed assets and employment for the treatment group and control group in each year of the sample period. The definition of the treatment and control group is based on the EUR 750m revenue threshold (see Section4). Panels B and D show the coefficient $\hat{\theta}_\tau$ on $\text{Treated}_i \times \text{Year}_t$ for log changes for tangible fixed assets and employment, as defined in Equation (1). The vertical lines indicate 95 percent confidence intervals for the coefficient estimates. In all panels, the dashed red lines indicate the announcement of the GMT in 2021, the solid red lines indicate the implementation of the GMT in 2024. Outcome variables are normalised to zero in 2018.

Figure 8: Treatment effects on subsidiary counts

Panel A: Log change total number of subsidiaries



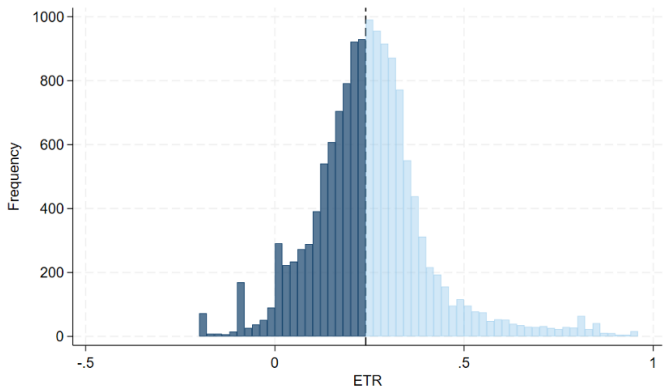
Panel B: Log change investment hub subsidiaries



Note: The charts show the coefficients $\hat{\theta}_\tau$ on $\text{Treated}_i \times \text{Year}_t$ as defined in Equation (1) for the log change in the total number of subsidiaries of an MNE group (Panel A) and the log change in the number of subsidiaries located in investment hubs (Panel B). The vertical lines indicate 95 percent confidence intervals for the coefficient estimates. The dashed red lines in both panels indicate the announcement of the GMT in 2021, the solid red lines indicate the implementation of the GMT in 2024. Outcome variables are normalised to zero in 2018. The definition of investment hubs follows Hugger et al. (2024).

Appendix

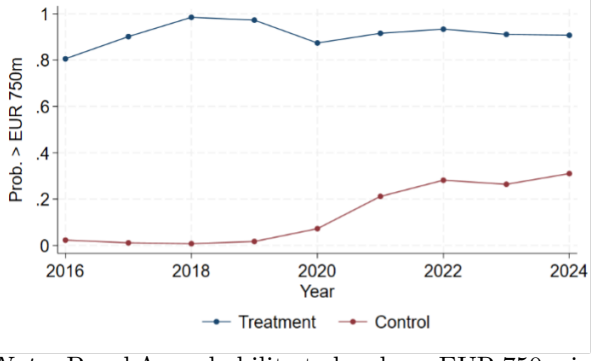
Figure A.1: Distribution of consolidated ETRs of MNEs



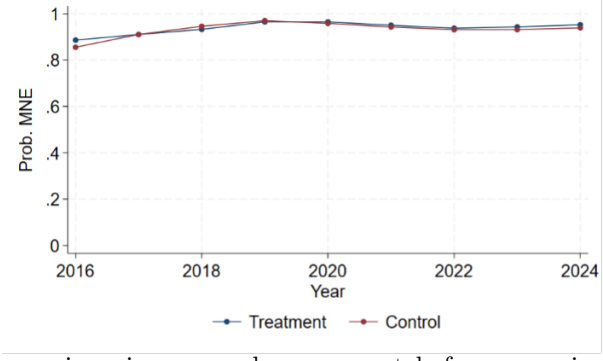
Note: Distribution of ETRs for MNEs with revenues between EUR 300m and 650m or between EUR 800m and 2,000m which could be selected for the treatment and control group. The underlying ETR distribution is based on fiscal years 2017–2019. The distribution shown is limited at −0.5 at the bottom and 1.0 at the top. The vertical dashed line indicates the median which is used as a cutoff to define the baseline sample.

Figure A.2: Assignment into treatment and control

Panel A: Probability of above EUR 750m revenues for treatment and control groups

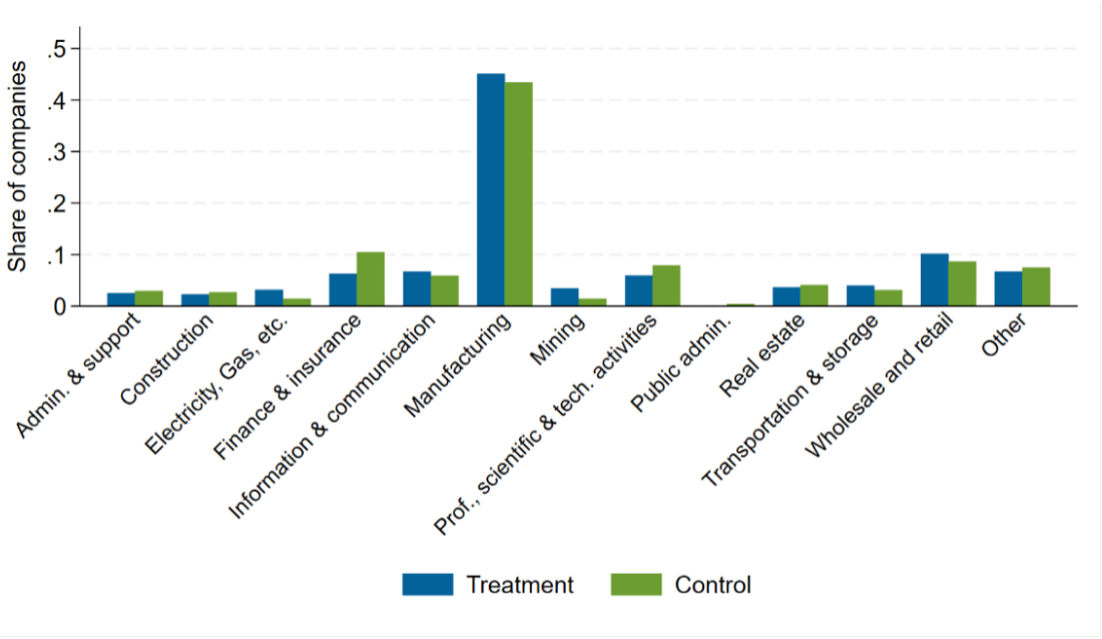


Panel B: Probability of MNE status for treatment and control groups



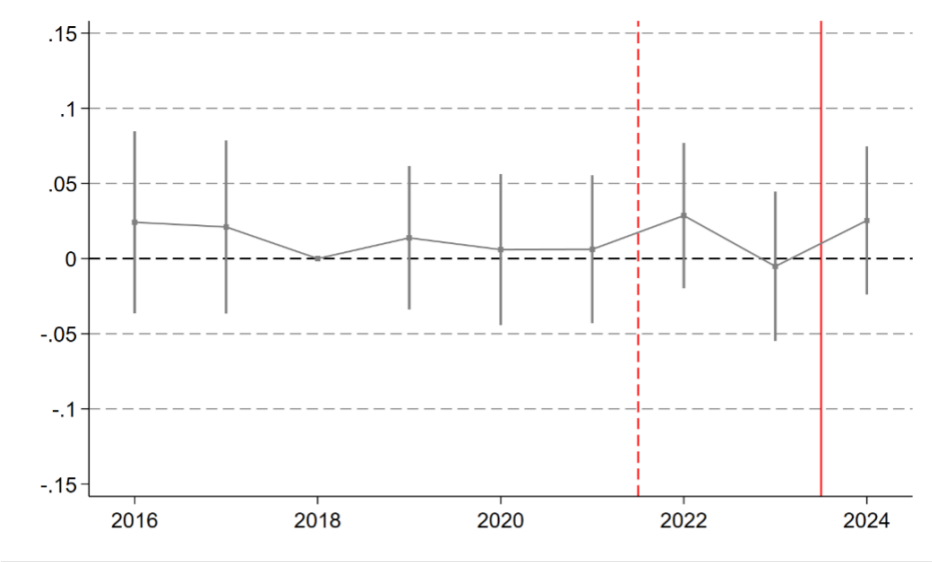
Note: Panel A: probability to be above EUR 750m in revenues in a given year, shown separately for companies in the treatment group (blue) and control group (red). Assignment is based on average revenues in the 2017–2019 period. Panel B: probability to have at least one foreign subsidiary recorded in Orbis in a given year, shown separately for companies in the treatment group (blue) and control group (red). Assignment is based on the record of at least one foreign subsidiary in Orbis in the 2017–2019 period.

Figure A.3: Distribution of treatment and control across industries



Note: Distribution of unique MNEs in the treatment and control group of the baseline sample across NACE codes. Industries accounting for less than 2% of the companies in both groups are pooled into "Other".

Figure A.4: Treatment effect for investment in intangibles



Note: The chart shows the coefficients $\hat{\theta}_\tau$ on $\text{Treated}_i \times \text{Year}_t$ as defined in Equation (1) for the log change in intangible fixed assets. The vertical lines indicate 95 percent confidence intervals for the coefficient estimates. The dashed red lines indicate the announcement of the GMT in 2021, the solid red lines indicate the implementation of the GMT in 2024. Outcome variables are normalised to zero in 2018.

Table A.1: Additional regression results

	(1) Taxation	(2) Pre-tax profits	(3) Investment in intangibles	(4) Total number of subs.	(5) Number of inv. hub subs.	(6) Taxation	(7) Pre-tax profits	(8) Investment in intangibles	(9) Total number of subs.	(10) Number of inv. hub subs.
Treated × 2020	0.0280	−0.0123	0.00598	−0.0127	−0.00918					
	(0.0601)	(0.0433)	(0.0256)	(0.0170)	(0.0172)					
Treated × 2021	0.0447	0.0496	0.00616	−0.0101	0.0112					
	(0.0596)	(0.0428)	(0.0251)	(0.0176)	(0.0179)					
Treated × 2022	−0.0583	0.00510	0.0286	−0.000993	0.0122					
	(0.0574)	(0.0429)	(0.0246)	(0.0177)	(0.0176)					
Treated × 2023	0.00115	0.0249	−0.00515	−0.00717	−0.00991					
	(0.0610)	(0.0445)	(0.0254)	(0.0170)	(0.0172)					
Treated × 2024	0.131**	0.0283	0.0254	0.00237	−0.00351					
	(0.0564)	(0.0433)	(0.0252)	(0.0174)	(0.0196)					
Treated × post-Announcement						−0.0522	−0.00483	0.000659	0.00481	−0.00346
						(0.0345)	(0.0287)	(0.0144)	(0.00699)	(0.00887)
Treated × post-Implementation						0.108**	0.00885	0.0140	0.0112	−0.00825
						(0.0437)	(0.0373)	(0.0179)	(0.00929)	(0.0141)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year and company fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pre-reform treated mean	0.110	0.0944	0.0788	0.0944	0.0416	0.110	0.0944	0.0788	0.0944	0.0416
Observations (firm-years)	10,705	11,293	13,588	14,624	9,504	10,705	11,293	13,588	14,624	9,504

Note: Columns (1)–(5) report estimated coefficients $\hat{\theta}_\tau$ where $\tau \in \{2016, \ldots, 2024\}$ from Equation (1). Columns (6)–(10) report the estimated coefficients $\hat{\theta}^A$ and $\hat{\theta}^I$ from Equation (2). The dummy variable Treated_i takes the value of 1 for companies expected to be in scope of the GMT. Dependent variables are measured as log changes. All specifications include year and company fixed effects as well as additional controls following Equations (1) and (2). Standard errors are clustered at the company level and are reported in parenthesis.

Table A.2: Robustness checks – Investment and employment

Panel A: Investment

	(1) Baseline	(2) Incl. higher taxed MNEs	(3) Incl. financial companies	(4) Excl. US MNEs	(5) Balanced sample	(6) Add. control: inflation	(7) Add. control: currency	(8) Add. control: profitability	(9) Add. control: lagged revenue	(10) Add. control: HQ x year	(11) Narrow revenue bounds	(12) No upper revenue bound	(13) Alternative donut hole	(14) Two out of four criterion
Treated × post-Announcement	0.00487	0.00553	0.00327	−0.000372	0.00635	0.00740	0.0115	0.00508	0.00464	0.00823	0.00260	0.00478	0.00627	−0.00451
	(0.00758)	(0.00402)	(0.00734)	(0.00778)	(0.00863)	(0.00791)	(0.00757)	(0.00749)	(0.00761)	(0.00821)	(0.0131)	(0.00628)	(0.00900)	(0.00646)
Treated × post-Implementation	0.00832	0.00687	0.00576	0.00529	0.0125	0.00833	0.0115	0.00744	0.00841	0.00874	0.0159	0.0173**	0.00593	0.00471
	(0.00988)	(0.00539)	(0.00954)	(0.0103)	(0.0108)	(0.0104)	(0.00985)	(0.00970)	(0.00985)	(0.0115)	(0.0166)	(0.00826)	(0.0116)	(0.00895)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year and company FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations (firm-years)	14,592	45,201	15,828	13,282	10,097	13,724	14,592	14,557	14,561	14,655	6,629	21,661	11,228	17,998

Panel B: Employment

	(1) Baseline	(2) Incl. higher taxed MNEs	(3) Incl. financial companies	(4) Excl. US MNEs	(5) Balanced sample	(6) Add. control: inflation	(7) Add. control: currency	(8) Add. control: profitability	(9) Add. control: lagged revenue	(10) Add. control: HQ x year	(11) Narrow revenue bounds	(12) No upper revenue bound	(13) Alternative donut hole	(14) Two out of four criterion
Treated × post-Announcement	0.00170	0.00338	0.00367	0.00137	−0.000811	0.00188	0.00189	0.000798	0.00200	0.00102	0.00472	0.00285	−0.00381	−0.00686
	(0.00616)	(0.00305)	(0.00564)	(0.00649)	(0.00714)	(0.00618)	(0.00637)	(0.00600)	(0.00611)	(0.00623)	(0.0110)	(0.00517)	(0.00728)	(0.00554)
Treated × post-Implementation	0.00243	0.00593	0.00406	0.00196	0.00600	0.00244	0.00310	0.000570	0.00181	0.00165	−0.00962	0.00900	−0.00531	−0.00688
	(0.00830)	(0.00417)	(0.00769)	(0.00910)	(0.00893)	(0.00836)	(0.00855)	(0.00817)	(0.00827)	(0.00866)	(0.0153)	(0.00672)	(0.00955)	(0.00734)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year and company FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations (firm-years)	10,940	33,969	9,126	9,586	11,974	10,884	10,940	10,907	10,910	10,998	4,911	16,776	8,504	14,086

Note: The different columns report estimated coefficients $\hat{\theta}^A$ and $\hat{\theta}^I$ from Equation (2) for a number of alternative samples. Some estimations include additional or alternative control variables as indicated (Columns (6)–(10)). The dummy variable Treated_i takes the value of 1 for companies expected to be in scope of the GMT. The dependent variables are the log change in tangible fixed assets (Panel A), and the log change in employment (Panel B). All specifications include year and company fixed effects as well as additional controls following Equation (2). Standard errors are clustered at the company level and are reported in parenthesis.

Table A.3: Placebo tests – Investment and employment

	Investment			Employment		
	(1) Baseline	(2) Non-MNEs	(3) Non-low ETR	(4) Baseline	(5) Non-MNEs	(6) Non-low ETR
Treated × post-Announcement	0.00487	−0.00627	0.00516	0.00170	0.00322	0.00371
	(0.00758)	(0.0137)	(0.00471)	(0.00616)	(0.00946)	(0.00337)
Treated × post-Implementation	0.00832	−0.0112	0.00558	0.00243	0.00532	0.00712
	(0.00988)	(0.0167)	(0.00634)	(0.00830)	(0.0123)	(0.00469)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year and company fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations (firm-years)	14,592	7,132	30,609	10,940	5,119	23,029

Note: Columns (1) to (6) report the estimated coefficients $\hat{\theta}^A$ and $\hat{\theta}^I$ from Equation (2). The dummy variable Treated$_i$ takes the value of 1 for companies expected to be in scope of the GMT. The dependent variables are the log changes in tangible fixed assets (columns (1)–(3)) and the log change in employment (columns (4) to (6)) at the MNE group level. All specifications include year and company fixed effects as well as additional controls following Equation (2). Standard errors are clustered at the company level and are reported in parenthesis.

Table A.4: Heterogeneity estimates – Investment and employment

Panel A: Investment

	(1)	Industry by profit-shifting concentration		Intangible fixed assets over total assets		Profit over tangible fixed assets	
	Baseline	(2) Low	(3) High	(4) Bottom quartile	(5) Top quartile	(6) Bottom quartile	(7) Top quartile
Treated × post-Announcement	0.00487	0.0117	−0.0013	0.00168	0.0162	−0.00503	0.00711
	(0.00758)	(0.00883)	(0.0140)	(0.0157)	(0.0180)	(0.0138)	(0.0197)
Treated × post-Implementation	0.00832	0.00875	0.00775	0.0365	−0.0115	0.00801	0.0290
	(0.00988)	(0.0120)	(0.0176)	(0.0225)	(0.0223)	(0.0200)	(0.0235)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year and company fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations (firm-years)	14,592	10,137	4,267	3,555	3,648	3,578	3,623

Panel B: Employment

	(1)	Industry by profit-shifting concentration		Intangible fixed assets over total assets		Profit over tangible fixed assets	
	Baseline	(2) Low	(3) High	(4) Bottom quartile	(5) Top quartile	(6) Bottom quartile	(7) Top quartile
Treated × post-Announcement	0.00170	0.00558	−0.00914	−0.0146	0.00430	−0.00620	0.0168
	(0.00616)	(0.00773)	(0.0105)	(0.0129)	(0.0136)	(0.0136)	(0.0146)
Treated × post-Implementation	0.00243	0.00211	−0.00168	0.0144	0.0419	−0.0192	0.0243
	(0.00830)	(0.0103)	(0.0141)	(0.0150)	(0.0182)	(0.0182)	(0.0186)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year and company fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations (firm-years)	10,940	7,651	3,136	2,708	2,680	2,694	2,702

Note: The different columns report estimated coefficients $\hat{\theta}^A$ and $\hat{\theta}^I$ from Equation (2) for a number of alternative sample splits. The dummy variable Treated_i takes the value of 1 for companies expected to be in scope of the GMT. All specifications include year and company fixed effects as well as additional controls following Equation (2). Standard errors are clustered at the company level and are reported in parenthesis.

Table A.5: Heterogeneity estimates – Size and IIR status – Investment and employment

	Investment (1) Split of treatment group by size	Investment (2) Split of treatment group by size & no upper bound	Employment (3) Split of treatment group by size	Employment (4) Split of treatment group by size & no upper bound		Investment (5) Split of treatment group by IIR implementation	Employment (6) Split of treatment group by IIR implementation
Treated × post-Announcement							
Below median treated baseline sample	0.00639	0.00563	0.00002	−0.00233	IIR headquarter	0.00526	0.0141*
	(0.00953)	(0.00951)	(0.00766)	(0.00758)		(0.0108)	(0.00797)
Above median treated baseline sample	0.00337	0.00117	0.00325	0.00209	Non-IIR headquarter	0.00487	−0.0063
	(0.00930)	(0.00916)	(0.00727)	(0.00714)		(0.00875)	(0.00758)
Above EUR 2,000m		0.00932		0.00586			
		(0.00731)		(0.00598)			
Treated × post-Implementation							
Below median treated baseline sample	0.00506	0.00366	0.00297	0.00286	IIR headquarter	0.00643	0.0169
	(0.0124)	(0.0123)	(0.0103)	(0.0102)		(0.0140)	(0.0115)
Above median treated baseline sample	0.0116	0.00648	0.00186	0.00010	Non-IIR headquarter	0.0129	−0.00736
	(0.0120)	(0.0119)	(0.0102)	(0.0100)		(0.0112)	(0.0101)
Above EUR 2,000m		0.0301***		0.0157***			
		(0.00965)		(0.00754)			
Controls	Yes	Yes	Yes	Yes		Yes	Yes
Year and company fixed effects	Yes	Yes	Yes	Yes		Yes	Yes
Pre-reform treated mean	0.085	0.081	0.042	0.036		0.081	0.042
Observations (firm-years)	14,592	21,984	10,940	17,009		14,492	10,940

Note: The different columns report estimated coefficients $\hat{\theta}^A$ and $\hat{\theta}^I$ from Equation (2). Columns (1) and (3) report coefficients separately for firms split by size relative to the median within the treated group. Columns (2) and (4) extend the sample by including firms with revenues above EUR 2,000m and report separate effects for this group. Columns (5) and (6) report coefficients separately for firms from headquarter jurisdictions with and without IIR. The dummy variable Treated_i takes the value of 1 for companies expected to be in scope of the GMT. All specifications include year and company fixed effects as well as additional controls following Equation (2). Standard errors are clustered at the company level and are reported in parenthesis.